Green Talk vs. Green Walk: How the Intention-Action Gap Shapes the Cost of Carbon in U.S. Loan Markets

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Abstract

This paper examines the effect of direct carbon emissions intensity on price and non-price terms of syndicated loans, and whether this relationship is moderated by the gap between stated environmental commitments and actual emissions performance (intention-action gap). This study uses a panel of 1,921 loans issued to U.S. listed companies between 2010 and 2025. The empirical research is based on OLS and logit regression models with industry and year fixed effects. The results show that an increase in carbon intensity is associated with significantly shorter maturities, across the full sample and specifically among less profitable firms, in the post-Paris period and for firms with stated emissions-reduction targets. No consistent effect is found on spread, amount, collateral, or covenants. The effect on maturity is further amplified for firms with intention-action gaps (greenwashers), a pattern that is robust across four distinct gap measures. The findings suggest that lenders manage carbon risk primarily through loan maturity, rather than price itself. This paper extends the carbon-risk pricing literature beyond spread. This study suggests that firms whose environmental commitments are not matched by actual performance are penalized by lenders.

Keywords: carbon intensity, scope 1 emissions, syndicated loans, loan maturity, greenwashing, intention-action gap

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1. Introduction

Environmental issues have been of particular significance in recent years, and have drawn the attention of government regulators, financial institutions, academics and the public globally. Climate fluctuations, greenhouse effects, abnormal weather conditions and air and water pollution, all raise concerns among society and policymakers. Many public and private organizations were created to tackle climate change and decrease pollution levels. The Intergovernmental Panel on Climate Change (IPCC) determined that current level of emissions would result in global warming, with global average temperature expected to reach 1.5 degrees higher than before industrial revolution between 2030 and 2052 (IPCC, 2018). This poses major challenges and risks to the planet and global economy. In 2015, United Nations Framework Convention on Climate Change (UNFCCC) introduced Paris Agreement (COP21), whose main objective is to limit "the increase in the global average temperature to well below 2 degrees above pre-industrial levels", ideally, to 1.5 degrees. To achieve this, participating countries agreed that global greenhouse gases (GHG) emissions should reach their peak as soon as possible and start declining right after. Such an agreement requires cooperation of all economic actors from both private and public sectors and increases peer pressure to meet temperature and emissions targets (Reghezza et al., 2022). The transition into low-carbon economy will likely impact financially all kinds of industries, as companies are affected by changes in climate policy and climate-related technology (Ho and Wong, 2023).

The creation of Paris Agreement turned carbon into a financial vulnerability of companies, as firms face financial risks related to climate changes and policies (Ehlers et al., 2022). Those risks can be divided into physical and transition risks. Physical risks refer to the challenges arising from climate-related catastrophes such as wildfires, floods or hurricanes. Whereas transition risks arise from changes in environmental regulations, climate policies and technologies (Reghezza et al., 2022). As carbon dioxide accounts for around 75% of the human-generated GHG (IPCC, 2022), the reduction of its emissions is of particular significance to fight climate change.

This thesis concentrates on a firm-level transition risk resulting from carbon dioxide emissions. Carbon dioxide emissions thereby become a source of financial risk for companies, commonly denoted in literature as "carbon risk" (Bolton and Kacperczyk, 2021), as high-emissions companies face high financial risks when governments impose regulations to reduce carbon emissions.

A growing body of literature examines whether financial markets in fact price firms' carbon risk (Bolton and Kacperczyk, 2021; Ehlers et al., 2022; Ho and Wong, 2023). Bolton and Kacperczyk (2021) showed that stock markets price carbon in equities, as stocks of high-emissions companies earn higher returns, due to the demanded carbon risk premium by investors. Building on that, other studies analyze whether banks similarly price carbon risk in the syndicated loan markets. Banks have started pricing carbon risk into loan spreads especially after Paris Agreement (Ehlers et al., 2022; Ho and Wong, 2023; Hu et al., 2024). Ehlers et al. (2022) additionally find that only direct (scope 1) emissions are priced by lenders. Lenders respond to carbon risk by shortening maturity (Chen et al., 2021; Ho and Wong, 2023), reducing loan amount (Chen et al., 2021), reducing access to new loans (Ben-Nasr et al., 2025; Ding et al., 2023; Reghezza et al., 2022) and imposing collateral requirements (Chen et al., 2021) or covenants (Choy et al., 2024; Lee and Zakota, 2026).

A related, but separate, growing literature examines the effect of environmental disclosure of companies on carbon risk, and whether the stated targets are matched by the actual performance, or are just symbolic (Attig et al., 2025; Du, 2015; Taddeo et al., 2026). This phenomenon is commonly referred to in literature as greenwashing or symbolic environmentalism (Delmas and Burbano, 2011; Yu, Luu and Chen, 2020). This stream of literature shows that voluntary disclosure and transparency can in fact lower the cost of capital (Ge et al., 2025; Lin et al., 2025; Matsumura et al., 2014; Pástor et al., 2022). However, the disclosure can act more as a symbolic tool to only pronounce environmental initiative rather than real actions (Delmas and Burbano, 2011; Du, 2015).

Nevertheless, the evidence on whether banks price real environmental actions over symbolic green labels remains limited (Cao et al., 2025; Ehlers et al., 2022; Kim et al., 2025; Sun and Yu, 2024). This study is at the intersection of those two literature streams, as it examines not only whether carbon impacts the structure of the loans, but also whether this effect is sensitive to credibility of firms' environmental commitments.

Three gaps remain in existing literature. First, most papers on carbon risk focus on its effect on loan spread, with less attention paid to other contractual terms. Second, studies rarely isolate direct (scope 1) emissions from total emissions, despite evidence that scope 1 emissions are priced by banks (Ehlers et al., 2022). Third, while greenwashing is described conceptually, it is rarely tested as the moderator of the relationship between carbon emissions and different loan terms. This matters because if banks cannot distinguish symbolic environmentalism from actual actions, greenwashing companies may obtain more capital without reducing their emissions, which undermines the carbon risk pricing mechanism documented in the literature.

To answer these gaps, this study examines the effect of direct carbon emissions intensity on price and non-price terms of syndicated loans, and whether this relationship is moderated by the gap between firms' stated environmental commitments and their actual emissions levels. Therefore, the research question is stated as follows: *Does direct carbon emissions intensity of U.S. companies result in more costly and restrictive loan terms (spread, amount, maturity, collateral and covenants), and does this effect intensify when a firm's environmental claims are not matched by actual performance?*

To answer the research question, this study analyzes a sample of 1,921 syndicated loans issued to 425 U.S. companies between 2010 and 2025. The study examines the effect of carbon intensity on five loan terms: spread, amount, maturity, collateral and covenants. The effect on the three continuous terms is estimated through OLS models, while the effect on the remaining binary terms (collateral, covenants) is estimated with logit models. All models include industry and year fixed effects, with standard errors clustered at the firm level. Moreover, the moderating role of intention-action gap is estimated through interaction term between carbon intensity and the gap indicator.

The study is based on three key theories in the fields of finance, information asymmetry and greenwashing. First, delegated monitoring theory (Diamond, 1984) motivates the focus on bank loans since banks act as informed intermediaries with access to private information and information advantage over arm's-length investors. Second, the mathematical model proposed by Stiglitz and Weiss (1981) under imperfect information in loan markets provides a basis for banks to adjust non-price loan terms rather than spread alone when risk of default is hard to estimate. Finally, voluntary environmental disclosure is an example of signaling (Spence, 1973), where companies convey unobservable information about their environmental commitments through costly but credible actions. However, if the disclosed signal is not verifiable, banks may fail to distinguish committed companies from the ones only using environmental claims to improve their reputation, a phenomenon called greenwashing (Delmas and Burbano, 2011).

This study finds that an increase in direct carbon emissions intensity is associated with shorter maturities of the loans, both in the full sample specification and among specific subsets of companies. Shorter maturities are found among less profitable companies, in the post-Paris period, and for firms with stated emissions-reduction targets. Beyond maturity, limited evidence emerges for other loan terms within specific subsamples. An increase in carbon intensity is associated with lower spreads for low-profitability firms, and in the post-Paris period, smaller amounts for companies in brown sectors, and higher covenants probability for companies with stated emissions-reduction targets. No consistent effect is found for spread, amount, collateral and covenants across the full sample. The carbon intensity effect on maturity is further amplified for companies with intention-action gaps, robust across different gap indicators. Weaker and less consistent moderation is found for spread, loan amount and collateral. Carbon intensity is associated with higher spreads, smaller loans and lower probability of collateral inclusion for greenwashing firms. However, the effects are not robust to other gap measures.

This study therefore contributes to the growing body of research on sustainable finance and carbon risk. It extends the carbon risk pricing literature from spread only analysis to full loan design. Additionally, by following Ehlers et al. (2022), it isolates direct (scope 1) emissions specifically. Further, it contributes to greenwashing literature, as one of the first studies to examine the intention-action gap as a direct and quantified moderator of loan terms. Beyond academic contributions, it can also offer practical advice to lenders and regulators on whether current disclosure-based screening is sufficient in terms of loans design. Moreover, it provides information to companies considering how to approach environmental disclosure. The findings suggest that publicly stating environmental commitments is not sufficient to obtain favorable loan terms. If the target is not backed by a reduction in emissions, such firms face significantly shorter maturities than non-greenwashing companies.

The rest of the paper is organized as follows. Section 2 comprises a literature review and develops hypotheses. Section 3 introduces the data set and methodology. Section 4 discusses results. Conclusions and limitations are presented in Section 5.

2. Literature Review and Hypotheses Development

2.1 How banks price environmental risk

Banks are more conservative while lending money to companies with higher levels of emissions for several reasons (Chen et al., 2021). First, if a company is a poor environmental performer, it faces higher regulatory, compliance and litigation risks, as a firm might face fines, cleaning costs and sometimes profit loss due to damaged reputation. Therefore, such companies have higher probability of default, leading banks to charge higher loan rates as compensation for that risk. Second, lenders are also exposed to higher levels of litigation risk, as lending to companies with poor environmental performance makes debtholders partially responsible for legal obligations (Chen et al., 2021). Finally, lending capital to companies with high emissions level may additionally hurt bank's reputation as it conflicts with social norms and interests. Therefore, banks incorporate environmental performance and responsibility into their risk and credit assessments. Additionally, many banks adopt Equator Principles, which require them to monitor environmental and social impacts of projects they finance (Power and Tandja-M., 2025).

2.2 From climate risk to firm-level carbon emissions

2.2.1 Climate risk: physical and transition risks

The environmental factors which banks price range from broad, external forces beyond firm's control (environmental regulation and exposure to climate risk) to firm's own emissions. The broadest of these factors is climate risk, which can be divided into physical and transition risks, as mentioned in Introduction. Several studies have determined the influence of physical risk on loans (Javadi and Masum, 2021; Li and Wu, 2023; Maztoul, 2025; Pavlič et al., 2026). Javadi and Masum (2021) in their study on drought effects (physical risk) showed that droughts raise U.S. loan spreads, but the effect is significant only for long-term loans of poorly rated firms. Another study of newly industrialized countries demonstrated that vulnerability of companies to physical climate risk has negative impact on bank lending, as banks cut the supply of loans (Maztoul, 2025). Consistent with those findings, Pavlič et al. (2026) found that banks tightened the lending conditions after the flood for exposed firms in Slovenia. Li and Wu (2023) using Germanwatch Climate Risk Index (measure of exposure to extreme weather events and natural disasters) also showed that higher climate risk reduced bank loan supply in China. Transition risk operates largely through environmental regulations and policy shifts. So far, the studies have been mostly focused on China, with some evidence from U.S. and Europe. Most findings point in the same direction, showing that stricter regulation and enforcement raise borrowing costs or restrict loans for high emitters. However, one form of regulation can work the other way, as climate-risk disclosure requirements may in fact lower the borrowing cost by reducing information asymmetry (Ge et al., 2025). Fard et al. (2020) have found in cross-countries analysis that companies subject to stricter environmental regulations face higher loan costs. They argue that those regulations impose more environmental liabilities on companies, which may result in substantial outflow of funds, resulting ultimately in lawsuits or bankruptcy. Fan et al. (2021) found evidence that when green loan regulations are strengthened, banks charge higher loan rates for non-compliant firms. However, both sides, companies and banks, would take into account environmental regulations only if they think that noncompliance will result in some adverse consequences, such as fines or lawsuits (Wu et al., 2023). Thus, stronger enforcement results in lower access to loans for companies (Wu et al., 2023), and fewer loans for heavily polluting firms after implementation of the credit rating policy, such as the Environmental Credit Rating Policy (ECRP) (Ding et al., 2025). Additionally, Choy et al. (2024) found that high environmental regulatory enforcement pushes lenders more to introduce environmental covenants in the loan terms, especially on real-property collateral. Another study from China found that carbon emissions trading results in companies from pilot regions getting fewer bank loans (Yu et al., 2022). What is more, banks charge higher risk premium to firms from locations with high carbon emissions after introduction of Paris Agreement (Hu et al., 2024). As noted, not all regulations raise costs. Ge et al. (2025) showed that although climate risk in general increases loan spreads, climate-risk disclosure mitigates this penalty. In particular, they demonstrated that introduction of SEC climate risk disclosure guidance reduced U.S. borrowers' costs by improving transparency and reducing information asymmetry. What is more, Reghezza et al. (2022), in their study of the European market found that after introduction of Paris Agreement, banks' loan share to high polluting firms decreased. Environmental regulation raises the probability and size of future liabilities. Li, Wen and Yang (2025) found that banks respond only to transition risks of companies, as those reduce firm profitability and increase the probability of default. Nevertheless, Bruno and Lombini (2023) found mixed results on the impact of climate transition risk on pricing and supply of syndicated loans. They found that banks charge higher margins to polluting firms after Paris Agreement, however they found no significant effect on supply measures. They showed that the relationship between lending variables and CO_2 emissions is nonlinear and amplified when Paris Agreement and environmental awareness are combined. Those findings may appear inconsistent, however it is important to distinguish that transition risk is priced more generally in spreads, whereas physical risk impacts loans supply after weather events, and is stronger for highly exposed borrowers.

2.2.2 Firm-level carbon emissions

Narrowing down from the external environment to the firm's actions, a firm's carbon dioxide emissions raise its credit risk directly, as found by previous studies (Capasso et al., 2020; Safiullah et al., 2021). High emitters are exposed to transition and compliance costs, which result in decrease in cash flows, reduced profitability and higher earnings volatility (Zhu and Zhao, 2022). What is more, companies with carbon-heavy operations face increased operational risks and tightened liquidity constraints, which increases default risk (Zhou et al., 2025). Banks also rely on information disclosed by companies and external credit ratings to assess the credit risk of a given company. Since disclosure is voluntary and imperfect, it creates information asymmetry, which makes it more difficult for lenders to assess high-polluting companies, which is reflected in poor accounting information quality and weaker credit ratings (Ding et al., 2023). Taken together, Ben-Nasr et al. (2025) found that companies with higher carbon risk have a lower share of bank debt, which is driven by both sides, as such firms want to avoid bank scrutiny (demand-side) and banks avoid lending to such companies (supply-side). This is consistent with the capital markets pricing carbon, as shown for equity markets by Bolton and Kacperczyk (2021), indicating the same carbon risk is recognized in both equity and debt markets. Bolton and Kacperczyk (2021) use absolute level of emissions or changes in emissions as proxy for carbon risk, whereas this study uses carbon intensity, a key component of a firm's environmental performance, as used in previous research (Ben-Nasr et al., 2025; Ehlers et al., 2022; Zhou et al., 2025; Zhu and Zhao, 2022).

Both the external climate and regulatory environment, and the firm's own carbon emissions lead to lower profitability, which raises probability of default. Additionally, incomplete emissions disclosure creates information asymmetry with advantage for borrowers. Since lenders' objective is to maximize the expected return from the loans, they adjust the loan terms to protect themselves. Banks respond to both margins, price (spread) and non-price conditions (e.g., amount, maturity, collateral, covenants). This is why this study expects that carbon-intensive borrowers will face worse loan terms, which the next section develops into hypotheses.

2.3 Carbon intensity and loan terms

The most direct response to carbon risk incurred by companies is for banks to price it into the loan spread (measured here as the all-in-spread-drawn, AISD; defined in Section 3.3.1), which compensates for the expected loss and provides a risk premium. Higher carbon intensity raises the expected losses incurred by the bank and the uncertainty of future cashflows and regulatory costs. Therefore, to protect themselves, lenders raise the cost of borrowing – interest rate of the loan. Carbon premium can be either interpreted as a compensation for expected loss, or as a broader risk premium resulting from, for example, damaged reputation of the bank when lending to a high polluting company. Carbon risk has a positive and significant relationship with the cost of loans (Zhou et al., 2025; Zhu and Zhao, 2022). Additionally, previous studies found that higher climate risk leads to higher loan prices (Ge et al., 2025; Li, Wen and Yang, 2025; Umar et al., 2025), with transition risk particularly raising spreads (Ho and Wong, 2023) and physical risk priced selectively for long-term loans (Javadi and Masum, 2021). Carbon intensity of direct (scope 1) emissions itself results in introduction of risk premium to the loans, after Paris Agreement (Ehlers et al., 2022). Chen et al. (2021) found as well that banks demand significantly higher loan spreads from firms with high levels of toxic chemical emissions. However, most of the studies include either cross-country analyses, or datasets from China or emerging markets, creating a gap that this study addresses. Therefore, by analyzing U.S. private loans with firm-level carbon intensity, a key component of a firm's environmental performance, hypothesis H1a is proposed, as follows.

H1a: Higher carbon intensity is associated with higher loan spreads.

Beyond the price of the loans, banks may adjust non-price contract terms to protect themselves from probability of default of the highly polluting companies. One of the mechanisms is to limit the quantity of loans (smaller committed funds, lower access) for the high polluting companies. When risk and information asymmetry problems are severe, raising just the rate of the loan may lead to adverse selection, thus instead of only charging higher interest rates, lenders may ration quantity of the loan (Stiglitz and Weiss, 1981). Hasan et al. (2026) found empirical evidence that banks committed to climate-related disclosure reduce the size and the availability of the loans to high emitters. Carbon risk has a negative relationship with bank debt ratio of companies, showing that high polluting firms are less likely to secure a bank loan (Ben-Nasr et al., 2025). Companies with higher emissions are granted fewer bank loans, due to the raised potential risks of environmental violations (Ding et al., 2023). Additionally, in European markets, banks reallocated credit away from high polluting companies after introduction of

Paris Agreement, reducing their access to the loans (Reghezza et al., 2022). At the bank level, climate risk reduced the aggregate supply of loans (Li and Wu, 2023; Maztoul, 2025). Taking all into account, hypothesis H1b is proposed to determine the relationship between carbon intensity and loan amount.

H1b: Higher carbon intensity is associated with smaller loan amounts.

Another non-price term of the loan, which can be adjusted by banks, is the maturity of the tranche. Shortening maturity reduces long-term exposure and forces more frequent revaluation and repricing of the loan. Having a shorter tenor allows the lender to exit the contract quicker or reprice it as carbon risk evolves. Therefore, for companies with high exposure to those risks, banks can adjust the maturity to adapt the conditions to the loan more often, given the environmental performance of the company. However, shortening the tranche period protects the lender but shifts the refinancing risk to the vulnerable borrower. Ho and Wong (2023) showed empirically that after Paris Agreement, high transition risk resulted in shortened loan tenors. Moreover, non-financial companies from China with higher carbon emissions face shorter credit term structures (Ding et al., 2023). Chen et al. (2021) find that firms with higher amounts of toxic chemical releases, measured with Toxic Release Inventory (TRI) also face shorter loan maturities. Alternatively, rather than shortening maturity, lenders may rely on other monitoring tools such as covenants and collateral (Lee and Zakota, 2026). On the other hand, Hasan et al. (2026) found banks' commitment to climate-related disclosure has no significant effect on loan maturity. Nevertheless, the available evidence, while still limited and rarely focusing on carbon intensity, points towards shorter maturities for high-emission borrowers, which this study examines.

H1c: Higher carbon intensity is associated with shorter maturity of loans.

Two other non-pricing terms include existence of collateral and covenants in the loan contract. Collateral is introduced to the loan contracts to minimize the lender's financial risk, where borrowers pledge assets which the lender can sell to recover the loan if borrower defaults. Secured loan improves the expected recovery on high-default borrowers, such as high emitters. However, this prediction might be unclear, as the assets pledged as collateral may be exposed to environmental risk. Particularly, they are exposed to physical risk, as in case of physical events (e.g., flood or wildfire), tangible assets such as property or equipment can be destroyed or damaged. Such assets can also be devalued by the low-carbon transition, and lenders can be additionally responsible for the costs of cleaning up the contamination on the property pledged as collateral (Choy et al., 2024). These factors can make lenders more cautious about collaterals while lending to companies with high levels of emissions. Nevertheless, the loss-mitigation effect is expected to dominate, thus borrowers with high carbon intensity will be more likely to be required to pledge a collateral. Use of environmental covenants is positively associated with use of collateral in loan terms (Choy et al., 2024; Lee and Zakota, 2026). Moreover, collateral is also required for companies with high levels of toxic chemical releases (Chen et al., 2021) and high polluters exposed to transition risk after Paris Agreement (Ho and Wong, 2023). Hasan et al. (2026), by contrast, find no significant effect on collateral, which creates a gap that this research addresses. Therefore, this paper analyzes the association between carbon intensity and probability of collateral in the loan contract, with hypothesis H1d as follows.

H1d: Higher carbon intensity is associated with a higher probability of a loan being secured (backed by collateral).

Covenants are monitoring tools which are used by banks to impose requirements on the companies' environmental behavior and actions (Lee and Zakota, 2026). Covenants allow banks to control and discipline high polluting companies and further reduce information asymmetry and moral hazard. They are the most effective in reducing toxic chemical emissions when environmental enforcement is strong (Choy et al., 2024). Covenants are treated as a complementary tool to price and collateral. Lee and Zakota (2026) showed that loans with environmental covenants tend to have longer maturity, to be secured and are more likely to be used for companies with high exposure to environmental risk. This implies that covenants are used together with other monitoring tools, rather than standalone term. Additionally, lenders are more likely to use them for high-polluting companies (Chen et al., 2021; Choy et al., 2024). However, Hasan et al. (2026) found that banks' commitment to climate-related disclosure has no significant effect on the usage of financial covenants, though the number of environmental covenants increases. This is consistent with the key idea that lenders use environmental covenants, specifically to monitor high polluters. Moreover, covenants are rarely evaluated in a carbon context, which this study addresses. Therefore, this paper analyzes the association between carbon intensity and probability of financial covenants in the loan contract, with hypothesis H1e as follows.

H1e: Higher carbon intensity is associated with a higher probability of financial covenants being included in the loan contract.

Companies with poor environmental performance face worse loan terms and lower access to bank credit, due to their exposure to climate risk and resulting higher probability of default. Banks price environmental risk into the loans based on the information available, which rely heavily on voluntary disclosures or external evaluation by third parties. However, these evaluations can be biased due to lack of information, imperfect rating systems or methodologies (Yin et al., 2025). Companies self-report the levels of their emissions along with the targets and commitments, which may not always match the actual level. Therefore, the question remains whether the credibility of those claims influences the penalty in worse loan terms.

2.4 The intention-action gap

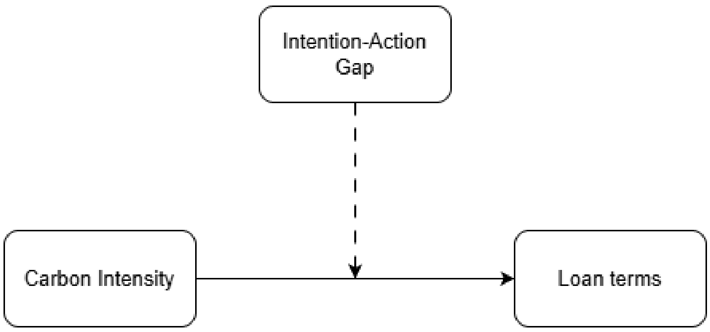
Voluntary disclosure of environmental commitments and actions is consistent with signaling theory (Spence, 1973), where firms convey unobservable information about their commitments through credible actions. Transparency reduces the information asymmetry priced by lenders, which makes companies seem like more credible borrowers. Carbon emissions have negative effect on the firm value, which is mitigated by voluntary disclosure (Matsumura et al. 2014). Similarly, costs of bank borrowing are mitigated through voluntary climate risk disclosure (Ge et al., 2025) or emissions disclosure (Lin et al., 2025). Companies that inform transparently about their environmental and social practices may be more attractive to investors (Pástor et al., 2022). However, the claims stated by firms might be symbolic, and do not reflect the actual emissions or actions (greenwashing). Many companies are engaged in greenwashing, misleading outside stakeholders about their real environmental performance (Delmas and Burbano, 2011; Du, 2015). Greenwashing occurs when there are differences between what companies disclose or commit to and what they actually do. This gap between stated environmental intentions and actual performance – what this study calls "the intention-action gap" – corresponds to the concept of greenwashing or symbolic environmentalism (Delmas & Burbano, 2011; Yu, Luu and Chen, 2020). Yu, Luu and Chen (2020) define "greenwashers" as firms with high level of transparent information on ESG data, but with poor ESG performance. Kim et al. (2025) show that under sustainability-linked loans (SLLs), high-transparency companies use them to certify themselves, whereas low-transparency borrowers use them only to greenwash. Nevertheless, the puzzle remains whether banks price real actions over empty labels. Ehlers et al. (2022) found that "green bank" label does not change the pricing of the loan. Sun and Yu (2024) found that Chinese companies with green credential (green Manufacturing certification - GMC) obtain more bank financing. Increase in cost of debt can be associated with selective environmental disclosure (greenwashing) (Attig et al., 2025) and misalignment between ESG disclosure and performance (Taddeo et al., 2026). Cao et al. (2025) find that borrowers with high ESG risks and low ESG performance face higher loan spreads. Additionally, the cost-saving effect of ESG performance (green "talk") depends on ESG risk (green "walk"), however not the other way round. That implies that companies with high disclosure may benefit from lower costs of borrowing, only if their ESG risk is also low. Therefore, if lenders price substance and penalize symbolic claims, then a company with a larger intention-action gap should face an additional penalty on top of the emissions effect. The hypothesis H2 is developed to test the moderating effect of the intention-action gap on the relationship between carbon intensity and loan terms.

H2: The positive association between carbon intensity and worse loan terms is stronger for firms with a larger intention–action gap.

Carbon intensity is predicted to worsen price (H1a) and non-price (H1b – H1e) terms of loans. This penalty is expected to be stronger with larger intention-action gap (H2), as shown in Figure 1.

Figure 1

Conceptual framework



Note. Carbon intensity is expected to be associated with less favorable loan terms. Specifically, firms with higher carbon intensity are predicted to face higher loan spreads (H1a), smaller loan amounts (H1b), shorter loan maturities (H1c), a higher likelihood of secured loans (H1d), and a higher likelihood of covenants (H1e). Furthermore, the intention–action gap is expected to moderate this relationship, such that the adverse effect of carbon intensity on loan terms is stronger for firms exhibiting a larger intention–action gap (H2).

Source: Own elaboration

3. Data and Methodology

3.1 Sample construction

The study uses an unbalanced panel of 1,921 private loan facilities issued to 425 listed U.S. firms over the period 2010-2025. The dataset is constructed by combining loan terms from DealScan with firm financials, emissions and ESG data from Refinitiv. The unit of analysis is the loan tranche. Coverage is strong for the key variables, referring to loan terms, carbon intensity and most controls. However, as some firm-level items used to construct moderating variables are not available for every company, the number of observations varies across specifications. Observations with missing values on any included variable in each model are thus excluded, as described in Section 3.4.

3.2 Empirical model

Following regression models are used to examine the relationship between carbon intensity and different loan terms, to analyze hypotheses H1a-H1e. Each loan term is examined with separate regression equations, with carbon intensity (CI) being the key regressor.

For the continuous variables (ln(spread), ln(amount) and ln(maturity)), the regression model is estimated by OLS, as follows.

$$Y_{ijt} = \alpha + \beta\, lnCI_{jt} + \gamma X_{jt} + \delta L_{ijt} + \mu_k + \tau_t + \varepsilon_{ijt} \qquad (1)$$

For the binary variables (secured and covenants), the regression model is estimated by logit, as follows.

$$\Pr(Y_{ijt} = 1) = \Lambda(\alpha + \beta\, \ln CI_{jt} + \gamma X_{jt} + \delta L_{ijt} + \mu_k + \tau_t) \qquad (2)$$

Y_{ijt} is the loan term for facility i of firm j in year t, $lnCI_{jt}$ is the natural logarithm of CI, X_{jt} is the vector of firm-level controls, L_{ijt} is the vector of loan-level controls (including the remaining loan terms) and Λ is the logistic cumulative distribution function. Industry (μ_k) and year (τ_t) fixed effects are included, as described in Section 3.3.3. Standard errors are clustered at the firm level. The coefficient of interest is denoted by β. All above used variables are defined in Section 3.3.

For the hypotheses H2, intention-action gaps are entered separately into the models, as an interaction with carbon intensity. The following equation presents the OLS regression model, as follows:

$$Y_{ijt} = \alpha + \beta_1 lnCI_{jt} + \beta_2 Gap_{jt} + \beta_3 \left(lnCI_{jt} \times Gap_{jt}\right) + \gamma X_{jt} + \delta L_{ijt} + \mu_k + \tau_t + \varepsilon_{ijt}, \quad (3)$$

where Gap_{jt} constitutes given intention-action gap. β_3 is the coefficient of interest as it evaluates whether the effect of carbon intensity on the loan terms depends on the firm's intention-action gap. The binary outcomes use the logit analogue of equation (3), with the same right-hand side inside $\Lambda\,(\cdot)$. All models are estimated and ran in R. Gaps measures are defined in Section 3.3.4.

3.3 Variables

This section defines the dependent, independent, control and moderating variables used in this study to assess defined hypotheses. Table 16 in Appendix provides detailed definitions of the variables. Before estimation, continuous variables were examined for potential outliers. Following common practice in the loan-pricing literature where continuous variables are winsorized to limit the effect of outliers (Chen et al., 2021; Hasan et al., 2026), extreme values were winsorized where necessary. Only variables with outliers were adjusted, leaving the remaining variables unchanged. Market-to-book was winsorized at the 5th and 95th percentiles and leverage at the 99th percentile. Additionally, some variables, namely spread, amount, maturity and number of lenders, are transformed using natural logarithm.

3.3.1 Dependent variables

This study uses five different dependent variables capturing the price and non-price loan terms, each mapping to one of the hypotheses developed in Section 2.3. The main dependent variable is the loan spread (*spread*), measured as the all-in-spread-drawn (AISD) in basis points, which represents the total fee over base rate paid by borrower on all drawn loan commitments. Following the loan pricing literature (Chen et al., 2021; Ehlers et al., 2022; Hasan et al., 2026; Javadi and Masum, 2021), the spread is entered into the regression as the natural logarithm of the all-in-spread-drawn to reduce right skewness. The remaining four variables capture non-price loan conditions. Loan amount (*amount*) corresponds to the committed funds available via individual tranche, and is entered in the form of natural logarithm, following previous studies (Chen et al., 2021; Ehlers et al., 2022; Ho and Wong, 2023; Javadi and Masum, 2021). Loan maturity (*maturity*) is the period in months between tranche active date and maturity date, also entered as a natural logarithm (Chen et al., 2021; Javadi and Masum, 2021). Collateral (*secured*) is a binary indicator equal to one if the tranche is secured and to zero otherwise. *Covenants* are also a binary indicator equal to one if the facility has any financial covenants, and zero otherwise. The spread, amount and maturity are continuous variables, whereas collateral and covenants are binary, resulting in different regression models as described in Section 3.2.

3.3.2 Independent variable

Following previous studies, the independent variable in this study is the carbon intensity (*CI*), which is a key indicator of the company's environmental performance, and prior studies often use it to capture firm's exposure to carbon risk (Ben-Nasr et al., 2025; Ehlers et al., 2022; Zhu and Zhao, 2022; Zhou et al., 2025). It is measured as corporate carbon emissions divided by sales revenue (Capasso et al., 2020; Chapple et al., 2013; Jung et al., 2018). This study follows Ehlers et al. (2022), who find that only scope 1 emissions are priced by banks. Therefore, carbon intensity is measured using only direct (scope 1) emissions. It is entered into the regression as the natural logarithm to reduce skewness (Zhou et al., 2025). The formula for calculating carbon intensity is as follows.

$$Carbon\ intensity\ (CI) = \frac{scope\ 1\ carbon\ dioxide\ emisions}{sales\ revenue} \tag{4}$$

The emissions data considered in the analysis are direct CO_2 emissions, which include direct carbon dioxide (CO_2) and CO_2 equivalents emissions, in tons, coming from sources that are owned or controlled by the company (scope 1 emissions). Rather than using absolute levels of carbon emissions (Bolton and Kacperczyk, 2021), this study uses the emissions scaled by sales, not to penalize big companies simply for being large. Therefore, the measure captures the firm's exposure to tightening carbon policy.

3.3.3 Control variables

The models in this study include firm-level and loan-level control variables drawn from the loan-pricing literature (Ben-Nasr et al., 2025; Chen et al., 2021; Javadi and Masum, 2021; Reghezza et al., 2022). Firm characteristics are obtained and calculated based on data from Refinitiv, and include firm's s*ize, leverage, tangibility, profitability* (Ben-Nasr et al., 2025), c*urrent ratio* (Reghezza et al., 2022), *market-to-book* (Chen et al., 2021; Javadi and Masum, 2021). *Size* is calculated as the natural logarithm of firm's total assets. *Leverage* is calculated as the ratio of total debt to total assets. *Tangibility* is determined as the proportion of tangible assets to total assets. *Profitability* is measured as return on assets (ROA), calculated as the ratio of net income to total assets (Lin et al., 2025). *Current ratio* is a ratio of current assets to current liabilities. *Market-to-book* is defined as the ratio of market value of equity of the company to its book value and is taken directly from the Refinitiv database. These controls capture the characteristics of the companies that are priced by the banks. Larger firms tend to have greater access to external financing, thus they are expected to pay lower spreads, whereas firms with high leverage have higher default risk and consequently higher cost of loans (Javadi and Masum, 2021). Firms with a higher tangibility ratio have greater value that can be pledged as collateral, which reduces the lender's risk. Such firms, for instance, are more likely to access bond markets (Ben-Nasr et al., 2025). In the loan market, this greater value is expected to reduce expected loss, and therefore the spread, all else equal. Similarly, profitable firms face lower risk of default, thus are expected to pay lower spreads (Javadi and Masum, 2021). The current ratio is included as the measure of liquidity, which determines the ability of the company to cover its short-term obligations with current assets (Reghezza et al., 2022). Greater liquidity is associated with lower short-term default risk and thus expected to be connected with lower borrowing costs. Lastly, companies with a higher market-to-book ratio have higher valuations and greater growth opportunities, which is expected to lead to lower interests on the loans (Javadi and Masum, 2021). The study also controls for loan characteristics in line with previous studies. *Tranche type and primary purpose* are categorical variables and enter as sets of dummy variables. *Tranche type* is divided into several groups: revolvers, main term-loan categories, bridge loan, and other loan types, with revolvers as the reference category, as shown in Table 4 in section 3.4. *Primary purpose* is grouped into general corporate purposes, takeovers, acquisitions and a residual category ("Other") for the remaining purposes (including working capital and spinoff), with general corporate purpose as reference category, as shown in Table 5 in section 3.4. The models also control for *number of lenders*, entered as a natural logarithm (Ho and Wong, 2023; Javadi and Masum, 2021) and *performance pricing* (Chen et al., 2021). Additionally, each model controls for the remaining loan terms, following the rule that whichever loan term is a dependent variable in a given model, is excluded from the controls in that model. Common macroeconomic conditions are absorbed by year fixed effects, therefore no separate macro controls are included. In addition, all models include industry fixed effects, constructed as dummy variables for SIC divisions, based on the corresponding range of two-digit SIC major groups (e.g., Manufacturing – major groups from 20 to 39), resulting in eight divisions, as described in Section 3.4.

3.3.4 Moderating variables

Moderating variable, *intention-action gap*, is introduced to determine whether banks penalize the discrepancy between a company's environmental claims and the real performance, which is known as symbolic environmentalism or greenwashing (Delmas & Burbano, 2011). Following

Yu, Luu and Chen (2020) who characterize greenwashers as companies with high environmental disclosure, but rather poor environmental performance, this study constructs four binary measures to quantify intention-action gap. Gap 1 is equal to one if the company has an emission-reduction target, but its direct carbon emissions intensity is above the median of the sector in that same year. This means that the stated target (intention) is not backed by real reduction of emissions (action). Gap 2 is equal to one if ESG score of the company is higher than median of the sector in that same year, but its carbon intensity is above the median of the sector in that same year. Gap 3 is equal to one if the company has Corporate Social Responsibility (CSR) committee, but its carbon intensity is above the median of the sector in that same year. Gap 4 is equal to one if the company supports the UN Sustainable Development Goal 13 (SDG 13) Climate Action, but it does not have senior executive's compensation linked to sustainability, CSR or health and safety targets.

3.4 Descriptive statistics

This section presents the summary statistics of the variables used in the analysis and describes the sample composition and coverage for each model. Additionally, it also reports the correlation between the variables and a comparison of loan-terms between high- and low-carbon borrowers.

Table 1 summarizes descriptive statistics for all continuous variables used in the models, including the main independent variable (carbon intensity, CI), loan characteristics and control variables. Several variables, including carbon intensity, loan amount and spread, follow a right-skewed distribution, as indicated by lower medians than means, and long upper tails. To reduce the skewness, these variables are entered into regressions in the form of natural logarithm. This transformation substantially reduces the skewness, as reflected in the table. Following standard loan-pricing literature, number of lenders and maturity are also entered in the form of natural logarithms. Therefore, Table 1 includes the summary statistics for both the original and transformed variables.

Table 1

Descriptive Statistics

Variable	Obs	Min	Mean	Med	Max	SD
CI	1,807	0.000	0.375	0.016	12.164	1.094
ln(CI)	1,806	-10.195	-3.679	-4.165	2.498	2.483
spread	1,921	1.750	163.365	135.000	1250.000	105.365
ln(spread)	1,921	0.560	4.956	4.905	7.131	0.528

Variable	Obs	Min	Mean	Med	Max	SD
amount	1,921	10.000	1829.974	1000.000	85000.000	3964.683
ln(amount)	1,921	2.303	6.822	6.908	11.350	1.124
maturity	1,760	2.000	44.293	60.000	243.000	26.650
ln(maturity)	1,760	0.693	3.554	4.094	5.493	0.765
numberoflenders	1,766	1.000	12.391	11.000	50.000	8.085
ln(lenders)	1,766	0.000	2.259	2.398	3.912	0.802
ta	1,910	226.704	36786.866	14990.000	551669.000	60484.835
size	1,910	12.331	16.581	16.523	20.128	1.304
leverage	1,910	0.000	0.355	0.339	0.795	0.158
tangibility	1,910	0.000	0.316	0.220	0.945	0.245
profitability	1,910	-0.348	0.050	0.041	0.631	0.074
curratio	1,919	0.079	1.502	1.347	7.045	0.794
mtb	1,916	-6.080	3.442	2.370	16.720	4.565
esg	1,921	9.550	62.069	63.980	93.120	14.782

Note. CI refers to direct carbon intensity. Spread is reported in basis points, amount and total assets are reported in millions of USD, maturity is reported in months. The number of observations varies across variables due to distinct levels of missing data. Full variable definitions are provided in Appendix [Table 16].
Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 2 presents the data coverage of all variables included in the analysis. The baseline sample equals 1807 observations, which is due to missing 114 observations for scope 1 carbon emissions. Loan terms variables are almost fully covered, with only the maturity variable lacking 161 observations. Most control variables are nearly covered, with 11 missing observations in most cases, due to missing data on total assets, and 155 observations missing for number of lenders control variable. Because these missing variables overlap substantially, the main models, H1a-H1e, are estimated on 1640 facilities.

Sample sizes for Gap 1, Gap 2 and Gap 3 are equal to 1629, as there are 11 sector-year cells containing only one firm, which are excluded from classification, as the median is then not a meaningful benchmark. Gap 4 is measured on the significantly smaller sample size (1374), as it is estimated using sdg13 variable which misses 630 observations. Accordingly, regression sample size varies by model specification, and the corresponding sample size for each model is reported in the results tables.

Table 2

Missing observations

Variable	Obs	Missing	% Missing
CI	1,807	114	5.9
spread	1,921	0	0.0
amount	1,921	0	0.0

Variable	Obs	Missing	% Missing
maturity	1,760	161	8.4
ta	1,910	11	0.6
size	1,910	11	0.6
leverage	1,910	11	0.6
tangibility	1,910	11	0.6
profitability	1,910	11	0.6
curratio	1,919	2	0.1
mtb	1,916	5	0.3
numberoflenders	1,766	155	8.1
secured	1,921	0	0.0
covenants	1,921	0	0.0
performancepricinggrid	1,921	0	0.0
tranchetype	1,921	0	0.0
purpose	1,921	0	0.0
target	1,921	0	0.0
esg	1,921	0	0.0
sdg13	1,291	630	32.8
excompensation	1,921	0	0.0
csr committee	1,890	31	1.6

Note. Percentages are calculated relative to the full sample of 1,921 loan facilities.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 3 summarizes the distribution of binary loan characteristics. The vast majority of loans in the sample are unsecured (77.4%). Financial covenants are included in 43.1% of the loans, while 40.5% of loan facilities contain a performance-based grid used to set the loan margin.

Table 3

Binary loan terms

Variable	%
Secured	22.6
Covenants	43.1
Performance pricing grid	40.5

Note. N = 1,921 loan facilities. Values represent the percentage of facilities for which the term equals "Yes."

Source: Author's own elaboration based on data from DealScan.

Tables 4 and 5 summarize the frequency distribution of categorical loan characteristics which are used as control variables in the model. For the tranche type variable, the reference category is the revolver type, which constitutes 41.1% of all facilities. The remaining categories constitute between 8.2% and 11.2% of all loans, with the other category equaling only 0.8%.

Table 4

Frequency distribution of tranche type

Tranche type	Obs	%
Revolver (ref)	790	41.1
Term Loan A	206	10.7
Institutional term loan	216	11.2
Term Loan (generic)	175	9.1
Delay-draw term loan	184	9.6
364-day facility	177	9.2
Bridge loan	158	8.2
Other	15	0.8

Note. N = 1,921 loan facilities.

Source: Author's own elaboration based on data from DealScan.

General corporate purpose is the reference category in the model for the purpose variable, as it accounts for the vast majority, 66.8% of all loans. The second most populated category is takeover purpose, with 16.9% frequency, whereas other purpose accounts for 4%.

Table 5

Frequency distribution of loan purpose

Loan purpose	Obs	%
General corporate (ref)	1284	66.8
Takeover	325	16.9
Acquisition	130	6.8
Working capital	56	2.9
Spinoff	50	2.6
Other	76	4.0

Note. N = 1,921 loan facilities.

Source: Author's own elaboration based on data from DealScan.

To deepen the analysis, the sample composition by division is also examined. Divisions are created by mapping the two-digit SIC major group to its corresponding division, as indicated by U.S. Department of Labor (OSHA, n.d.). The sample does not contain any loan taken by company from division A (Agriculture, Forestry and Fishing) or division H (Finance), therefore the analysis is made for remaining eight divisions. Table 6 summarizes the population of each division in terms of number of facilities and companies, as well as mean values of continuous variables by division, and additionally median of carbon intensity. The median is used to classify divisions as brown or clean, as described in section 4.1. Division D (Manufacturing) accounts for the majority of the facilities in the dataset (54.1 %). Two divisions, C (Construction) and J (Public Administration), include only six and one firm, respectively, and are therefore not interpreted individually. Carbon intensity is concentrated in Mining (0.69) and Transportation, Communications, Electric, Gas and Sanitary Services (1.48), which motivates the use of industry fixed effects in the models. This approach enables isolating the carbon effect on loan terms from the inherent characteristics of the industry. One company belonging to Public Administration division lacks observations for maturity and number of lenders, thus the means of those variables are not computable.

Table 6

Summary Statistics by Industry Division

Division	N Fac.	N Firms	CI mean	CI median	Spread	Amount	Maturity	TA	Leverage	Tangib.	Profit.	Curr. Ratio	Mtb	N Lenders
B Mining	85	24	0.69	0.33	243.15	1,506.59	41.65	23,765.98	0.26	0.67	0.06	1.70	1.63	12.99
C Construction	9	6	0.02	0.02	198.39	775.00	36.33	10,092.86	0.35	0.08	0.06	3.59	2.60	9.44
D Manufacturing	1,039	220	0.09	0.01	159.21	1,816.23	44.73	29,269.29	0.35	0.21	0.05	1.73	3.51	12.86
E Transportation, Communications, Electric, Gas and Sanitary Services	374	77	1.48	0.55	166.54	2,013.06	43.70	70,071.72	0.39	0.60	0.02	0.91	2.60	11.53
F Wholesale Trade	85	18	0.01	0.00	148.15	1,258.82	41.64	21,013.07	0.24	0.14	0.05	1.72	3.99	11.30
G Retail Trade	125	24	0.01	0.00	131.96	2,044.05	38.52	45,285.76	0.44	0.39	0.10	1.20	4.08	11.92
H Services	203	55	0.01	0.00	170.51	1,858.57	49.63	23,905.87	0.39	0.25	0.04	1.37	4.79	12.20
J Public Administration	1	1	0.00	0.00	175.00	600.00	N/A	5,382.30	0.45	0.39	0.03	0.67	3.52	N/A

Note. All values are reported as division means, except for CI where both mean and median are reported. N Fac. = number of loan facilities (tranches); N Firms = number of unique firms; N Lenders = average number of lenders per facility. CI = carbon intensity, measured as direct carbon emissions divided by sales revenue. CI, Spread, Amount, Maturity, Total Assets (TA) and Mtb are reported as the industry mean in the units of the original dataset (spread in basis points, amount and TA in millions, maturity in months). Leverage, Tangibility (Tangib.), Profitability (Profit.) and Current Ratio (Curr. Ratio) are expressed as ratios. Maturity and N Lenders are not reported for Division J, as data is unavailable for single facility.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 7 presents a correlation matrix between continuous and binary variables used in the regression models examining hypotheses H1a-H1e. Only one correlation is exactly at the threshold of 0.7 (tangibility × ln(CI), 0.71). Other than that, carbon intensity is weakly correlated with controls, thus variance inflation factor (VIF) analysis is conducted to confirm no multicollinearity. The VIF for carbon intensity is around 3.03, and the highest value across all continuous regressors is that for tangibility at 3.43, which is well below the common threshold of 5. This confirms that this correlation is not a multicollinearity concern. Categorical variables (tranche type, division) showed elevated nominal VIF values due to multiple degrees of freedom, but their Generalized Variance Inflation Factors (GVIF) values of 1.17 and 1.14, confirm no further multicollinearity.

Table 7

Correlation matrix among variables

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1.ln(CI)	–													
2. ln(Spread)	.16	–												
3. ln(Amount)	-.12	-.26	–											
4. ln(Maturity)	.01	.25	-.21	–										
5. ln(Lenders)	-.07	-.04	.28	.21	–									
6. Size	.04	-.34	.61	-.34	.24	–								
7. Leverage	.10	.11	.07	-.01	.01	.05	–							
8. Tangibility	.70	.12	-.11	.01	-.12	.10	.12	–						
9. Profitability	-.08	-.14	.04	.00	.01	-.12	-.03	-.07	–					
10. Current ratio	-.19	.10	-.12	.10	-.02	-.35	-.12	-.33	.06	–				
11. Market-to-book	-.17	-.14	.05	.01	.02	-.06	.03	-.11	.14	.01	–			
12. Secured	.07	.53	-.20	.35	-.08	-.32	.13	.02	-.04	.07	-.06	–		
13. Covenants	-.03	-.01	-.02	.07	.19	-.04	-.09	-.03	.03	.05	.03	-.05	–	
14. Performance pricing grid	-.12	-.13	.13	.02	.29	.11	-.12	-.12	.06	.03	.08	-.22	.50	–

Note. ln(CI) is the independent variable; ln(Spread), ln(Amount), ln(Maturity), Secured, and Covenants are dependent variables; Size, Leverage, Tangibility, Profitability, Current ratio, and Market-to-book are firm-level control variables; Lenders and Performance pricing grid are loan-level control variables. Significance levels are omitted from this table, full matrix is reported in Appendix [Table 17]

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 8 presents the results of difference-of-means test conducted with Welch's t-test, where firms were split at median of carbon intensity into two equally sized groups: high-carbon and low-carbon emitters. The table shows means for both groups, along with the differences and significance levels. Those results show unconditional differences, but do not control for any firm-level or loan-level variables, therefore are not a formal test of H1. The differences in means are significant for all variables, except for covenants. The majority of results coincide with the hypotheses as they show that high emitters obtain 20 % higher spreads (0.182 log points), 31.5 % lower amounts of loans (-0.378 log points), and loans for high emitters are by 9.9 percentage points more likely to be secured with collateral. However, the result for maturity is contradictory to the effect predicted in hypothesis H1c. The test shows that high-carbon firms obtain slightly longer maturities (6.6%, 0.064 log points). Nevertheless, maturity is strongly tied to loan type and firm's size and therefore once accounting for control variables, regression results may differ.

Table 8

Comparison of means for high- and low- carbon emitters

Variable	M High	M Low	Difference
ln(Spread)	5.043	4.861	0.182***
ln(Amount)	6.654	7.032	-0.378***
ln(Maturity)	3.580	3.516	0.064*
Secured	0.269	0.170	0.099***
Covenants	0.435	0.419	0.016

Note. High and Low refer to loans with above- and below-median carbon intensity, respectively. Difference = M High − M Low. Statistical significance is assessed using Welch's two-sample t-test.

* p < .10. **p < .05. ***p < .01

Source: Author's own elaboration based on data from DealScan and Refinitiv.

4. Results

4.1 Main results (H1a – H1e)

The OLS regression model is built to examine the effect of carbon intensity on spread, amount and maturity of loans obtained by companies in the dataset. For binary loan terms (collateral and covenants), a logit model is estimated, as described in Section 3.2. Table 9 presents the summary of results of the regressions for all the loan terms mentioned.

Table 9

H1a – H1e Regression Results

Variable	**Spread**	**Amount**	**Maturity**	**Collateral**	**Covenants**
ln(CI)	**0.004 (0.013)**	**-0.027 (0.019)**	**-0.019** (0.010)**	**0.085 (0.097)**	**-0.011 (0.068)**
Firm controls	Included	Included	Included	Included	Included
Other loan terms	Included	Included	Included	Included	Included
Loan controls	Included	Included	Included	Included	Included
Division FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.420	0.562	0.675	0.552	0.335
Pseudo R^2	0.353	0.270	0.486	0.540	0.273
BIC	1,973.6	4,035.8	2,280.7	1,026.3	1,966.1

Note. This table reports coefficients from regressions of loan-contract terms on ln(CI), with the full set of firm controls, other remaining loan terms, and loan controls included. Full coefficients for all controls are reported in Appendix [Table 18]. Standard errors, clustered at the firm level, are shown in parentheses. Models for Spread, Amount, and Maturity are estimated by OLS; Collateral and Covenants are estimated by logistic regression. R^2 values are reported for OLS models and pseudo-R^2 (McFadden) for logit models.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

For the loan spread, coefficient on carbon intensity is positive but statistically insignificant (0.004), which does not provide evidence that carbon intensity is associated with loan pricing in the main model. The sign is consistent with the predicted direction, however the hypothesis H1a is not supported. For the loan amount, the effect of carbon intensity remains insignificant and negative (-0.027), consistent with predicted direction, however hypothesis H1b cannot be supported. The only significant effect is noted for loan maturity (-0.019, $p < .05$), which indicates that companies with higher carbon intensity of direct emissions obtain loans with shorter maturities, thus hypothesis H1c is supported. For collateral, the coefficient on carbon intensity is positive, but not statistically significant (0.085), providing no evidence that carbon intensive firms are more likely to have their loans backed by collateral. Although the sign is positive as predicted, the hypothesis H1d is not supported. Lastly, for covenants, the coefficient is negative and statistically insignificant (-0.011), which contradicts the predicted sign of the relationship. There is no evidence that carbon intensity is associated with the use of financial covenants, and hypothesis H1e is not supported.

Since the pooled results show only limited evidence of a carbon-related effect on loan terms, the following subsample analysis examines more in-depth whether such effects are concentrated within specific groups. Five additional analyses are conducted. First, companies are split by size, as large companies may possess greater bargaining power and as a result may be less exposed to carbon-related loan effects. Second, the sample is split around the 2015 Paris Agreement, which has increased lenders' attention to climate risk (Ehlers et al., 2022; Reghezza et al., 2022). Third, firms are classified according to credit quality, determined through profitability, which follows study of Javadi and Masum (2021) who found that climate effect is concentrated in poorly rated firms. Fourth, sectors are divided into brown and clean categories, as carbon effects are expected to be more pronounced in high-emitting industries. Finally, firms are grouped based on whether they have set emissions-reduction target, which allow to analyze whether banks respond to stated climate commitments. Each model includes an interaction term between carbon intensity and the relevant group indicator to estimate whether carbon effect is different across groups. Table 10 shows the results of regressions within each subsample. The effect for reference group is given by the coefficient on carbon intensity (β_1), and for remaining group is calculated as the sum of the carbon intensity and interaction coefficients ($\beta_1 + \beta_3$). When the coefficient β_1 is not significant, it cannot be distinguished from zero, thus the effect for remaining group is reduced to the interaction coefficient β_3.

Table 10

Coefficient Estimates by Subsample

Subsample	Spread	Amount	Maturity	Collateral	Covenants
Size					
ln(CI) (small firms)	0.008	-0.013	-0.013	0.076	-0.051
ln(CI) × Large	-0.008	-0.030	-0.013	0.027	0.080
Post-Paris					
ln(CI) (pre-Paris)	0.028	-0.038	-0.0004	0.084	-0.068
ln(CI) × Post	-0.034**	0.015	-0.026**	0.002	0.081
Profitability					
ln(CI) (high profitability)	0.018	-0.012	-0.002	0.154	0.010
ln(CI) × Low profitability	-0.024**	-0.026	-0.029**	-0.105	-0.036

Subsample	Spread	Amount	Maturity	Collateral	Covenants
Brown/Clean					
ln(CI) (clean)	0.002	0.013	-0.015	-0.003	0.051
ln(CI) × Brown	-0.008	-0.081**	-0.019	0.167	-0.163
Target					
ln(CI) (no target)	0.013	-0.026	-0.005	0.076	-0.089
ln(CI) × Target	-0.017	0.002	-0.027**	0.034	0.138*

Note. ln(CI) is the natural logarithm of carbon intensity; interaction terms capture the differential effect for the named subgroup. All models include the full set of controls and fixed effects as in the main specification. Standard errors are omitted from this table. Full results, including standard errors, are reported in Appendix [Tables 19-23]. *p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

The first subsample is created by dividing companies into large and small, using the sample median as threshold. Neither baseline coefficient (small firms) nor interaction terms are significant for any of the analyzed five loan terms, in this subsample. This indicates that carbon intensity has no measurable effect on small firms in this model, and the effect does not differ significantly for large companies. Overall, these results provide no evidence that large firms might be shielded from carbon-related effects through greater bargaining power.

The second subsample splits loan tranches issued before and after 2015. In the pre-Paris sample, none of the carbon intensity coefficients are significant. The interaction term (ln(CI) × Post) is negative and significant for spread (-0.034, $p < .05$). Since baseline coefficient is treated as null, the effect for post-Paris subsample equals to interaction term itself, which indicates that higher carbon intensity is associated with lower spreads after 2015. This is contrary to theoretical expectations and current literature findings that carbon risk pricing intensified after Paris Agreement (Ehlers et al., 2022; Hu et al., 2024). However, this result needs to be interpreted with caution, as the pre-Paris subsample includes only 466 observations, compared with 1279 observations in the post-Paris subsample. Thus, the result may partially reflect broader post-Paris credit market conditions rather than the carbon effect. In addition, the interaction term for maturity (ln(CI) × Post) is negative and significant (-0.026, $p < .05$), suggesting that high-carbon firms receive significantly shorter maturities after 2015. This result is consistent with findings of Ho and Wong (2023), who show that high transition risk led to shorter maturities, specifically after 2015. The direction of effect is also in line with the prediction of hypothesis H1c. No significant interaction is found for amount, collateral or covenants.

The third subsample compares companies with high and low profitability, splitting the sample at the median. For companies with high profitability, none of the coefficients on carbon intensity are significant, thus the baseline is treated as zero. The interaction term (ln(CI) × Low profit) for spread is negative and statistically significant (-0.024, $p < .05$), indicating that carbon intensity is associated with lower spreads for companies with poor performance. This result, although contrary to the prediction that higher carbon intensity is associated with higher spreads, can be interpreted in terms of adverse selection. Stiglitz and Weiss's (1981) theoretical model shows that applying higher interest rates may in fact worsen adverse selection and moral hazard, consequently lowering bank's expected return. This provides a theoretical basis for lenders adjusting non-price loan terms. Thus, for poor performers, banks may avoid raising prices as the risk of default is increased. Additionally, this heterogeneity shows that the price effect occurs but only for given subsets of borrowers, which was hidden in the full sample. Moreover, the interaction term on maturity is also negative and statistically significant (-0.029, $p < .05$), highlighting that carbon intensity is associated with shorter maturities for low profitability firms. This is consistent with findings of Ho and Wong (2023) and Chen et al. (2021), that companies with high emissions obtain shorter maturities. Additionally, Zhu and Zhao (2022) show that carbon-intensive operations of companies lower the profitability and raise earnings volatility, which increases the risk of default of such firms. Thus, by shortening maturity for low profitability borrowers, banks may reduce their long-term exposure to financial risk and reevaluate the loan as the risk evolves.

The fourth subsample examines differences between brown and clean sectors. Sectors are divided as brown when the median of sector's direct carbon emissions is higher than overall sample median. This results in three sectors being classified as brown, namely Mining (B), Construction (C) and Transportation, Communications, Electric, Gas and Sanitary Services (E). In clean sectors, none of the estimated effects are significant statistically, so the baseline is treated as null. The interaction term (ln(CI) × Brown) reveals that only the difference in the effect on loan amount is statistically significant (-0.081, $p < .05$), indicating that carbon intensity is associated with smaller loans in brown sectors. This is consistent with logic behind the study of Ho and Wong (2023), who find that banks price-in climate transition risk for loans for companies from brown sectors, specifically after Paris Agreement. Additionally, they show that lenders apart from including risk premium may adjust the non-price loan terms for such companies, such as shortening maturity or imposing collateral requirement. The finding in this study on effect on loan amount extends the same sector-based logic to an additional non-price loan term. Overall, in brown sectors, high-emitting companies obtain significantly smaller loans, whereas in clean sectors carbon has no effect on loan terms.

Lastly, companies are distinguished based on whether they have stated emissions-reduction targets. For companies without target, no effects are significant, thus baseline is treated as null as well. The interaction term on covenants (ln(CI) × Target) shows that the difference between groups is significant and positive (0.138, $p < .10$). This suggests that carbon intensity is associated with significantly higher probability of obtaining covenants for companies with stated targets. Additionally, the interaction with maturity is significant and negative (-0.027, $p < .05$). This indicates that for companies with stated targets, carbon intensity is associated with shorter maturities. This pattern is consistent with the interpretation that banks may respond stronger when stated commitments do not match the actual decline of emissions. This is in line with broader concept of greenwashing or symbolic environmentalism (Delmas and Burbano, 2011; Yu, Luu and Chen, 2020), where larger intention-action gap may further strengthen the probability of covenants and shorten the maturity of the loan, which is further examined in Section 4.2.

Taken together, subsample results reveal that the effect of carbon intensity on loan terms is concentrated in subgroups, despite being limited in the pooled sample. The negative effect of carbon intensity on maturity is significant for low profitability firms, firms with stated targets and after Paris Agreement. Loan size is reduced primarily for high-emitting companies operating in brown sectors, and spread is reduced for low profitability companies. These results are also robust to the specification of the loan type and purpose controls. In this model the full set of tranche type and purpose dummies are replaced by single binary dummies (1 = revolver loans; 1 = general purpose). The coefficients on carbon intensity are similar in magnitude and significance across the two models, with one difference where loan amount coefficient loses its significance under simplified controls. The subsample analysis yields broadly the same conclusions under simplified model. Regression results from the simplified model can be found in Appendix (Tables 24-28). The findings that firms with a stated target and higher carbon intensity obtain shorter maturities suggest that banks may respond to the discrepancy between stated intentions and actual actions rather than solely stated commitment. This possibility is examined further in the following section, which assesses whether the gap between claimed intentions and actual carbon performance affects loan terms.

4.2 Moderation results (H2)

To estimate the moderating role of the intention-action gap, additional regression models are estimated by introducing four gap measures into the models separately as an interaction with carbon intensity, as specified in equation (3). As in the main analysis, OLS models are estimated for continuous variables and logit models for binary loan terms. The models are estimated for four specified gap measures (Gap 1, Gap 2, Gap 3 and Gap 4) and carbon intensity. In each model, the effect of carbon intensity for firms without a gap (non-greenwashers) is given by the coefficient on carbon intensity (β_1), and for firms with a gap (greenwashers) is calculated as the sum of the carbon intensity and interaction coefficients ($\beta_1 + \beta_3$). When both coefficients are significant with opposite signs, a linear restriction test is used to determine the combined effect. The interaction between carbon intensity and the gap captures whether the effect of carbon intensity on loan terms differs based on the firm's intention-action gap. Tables 11-15 present the main findings on the moderating role of each gap measure across five loan terms. To keep tables concise, only key coefficients are kept (carbon intensity, the gap indicator and their interaction). Full regression results for each gap indicator, including all control variables, are provided in Appendix (Tables 29-32).

Table 11

Moderating Effect of the Intention-Action Gaps on the Relationship Between Carbon Intensity and Loan Spread

Variable	Gap 1	Gap 2	Gap 3	Gap 4
ln(CI) β_1	0.011 (0.014)	0.008 (0.013)	0.001 (0.014)	0.002 (0.015)
Gap β_2	-0.078 (0.065)	-0.014 (0.049)	0.059 (0.054)	-0.023 (0.089)
ln(CI) × Gap β_3	**-0.011 (0.017)**	**0.019 (0.013)**	**0.032** (0.014)**	**0.006 (0.017)**
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,629	1,629	1,629	1,374
Pseudo R^2 / R^2	0.42219	0.42409	0.42595	0.42349

Note. This model is estimated by OLS. Controls include firm-level controls (size, leverage, tangibility, profitability, current ratio, market-to-book), other remaining loan terms, and loan-level controls (number of lenders, performance pricing grid, tranche type, purpose). Full coefficients, including all controls, are reported in Appendix [Tables 29 (Gap 1) through 32 (Gap 4)]. Standard errors, clustered at the firm level, are shown in parentheses.

*p < .10. ** p < .05. *** p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

The relationship between carbon intensity and loan spread is only moderated by Gap 3. The carbon intensity coefficient β_1 for non-greenwashers is not statistically significant and thus is attributable to zero. Interaction term is positive and significant (0.032, $p < .05$), therefore the effect for greenwashers resulting from the sum of both coefficients is attributable to just interaction term. Therefore, it can be stated that for firms with an intention-action gap, measured by Gap 3, increase in carbon intensity is associated with higher spreads. This is consistent with the predicted direction of the moderation, as intention-action gap further increases the spread of the loan. However, the effect is not replicated under other gap measures.

Table 12

Moderating Effect of the Intention-Action Gaps on the Relationship Between Carbon Intensity and Loan Amount

Variable	Gap 1	Gap 2	Gap 3	Gap 4
ln(CI) β_1	-0.019 (0.022)	-0.023 (0.018)	0.004 (0.021)	-0.022 (0.019)
Gap β_2	-0.107 (0.096)	-0.107 (0.123)	-0.213** (0.099)	0.268 (0.181)
ln(CI) × Gap β_3	**-0.028 (0.025)**	**-0.056* (0.032)**	**-0.034 (0.025)**	**0.051 (0.034)**
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,629	1,629	1,629	1,374
Pseudo R^2 / R^2	0.56992	0.57150	0.57077	0.58191

Note. This model is estimated by OLS. Controls include firm-level controls (size, leverage, tangibility, profitability, current ratio, market-to-book), other remaining loan terms, and loan-level controls (number of lenders, performance pricing grid, tranche type, purpose). Full coefficients, including all controls, are reported in Appendix [Tables 29 (Gap 1) through 32 (Gap 4)]. Standard errors, clustered at the firm level, are shown in parentheses.

*$p < .10$. ** $p < .05$. *** $p < .01$.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

For the loan amount, only Gap 2 moderates the relationship with carbon intensity. Since the effect for greenwashers is given by $\beta_1 + \beta_3$ and β_1 is not significantly different from zero, this effect is reduced to the interaction coefficient. Therefore, for companies with an intention-action gap, measured with Gap 2, increase in carbon intensity is associated with lower loan amounts, which is consistent with the predicted direction of the effect. However, similarly to loan spread, the effect is not robust across other specifications. Additionally, Gap 3 itself has a significant, negative effect on loan amount ($\beta_2 = -0.213$, $p < .05$), indicating that greenwashing firms obtain lower loan amounts, regardless of their carbon intensity. Nevertheless, the effect is not replicated across other measures.

Table 13

Moderating Effect of the Intention-Action Gaps on the Relationship Between Carbon Intensity and Loan Maturity

Variable	Gap 1	Gap 2	Gap 3	Gap 4
ln(CI) β_1	-0.017* (0.010)	-0.017* (0.010)	0.003 (0.011)	-0.008 (0.011)
Gap β_2	-0.085 (0.056)	-0.057 (0.052)	-0.184*** (0.062)	-0.223** (0.091)
ln(CI) × Gap β_3	**-0.048*** (0.013)**	**-0.030** (0.013)**	**-0.048*** (0.014)**	**-0.055*** (0.018)**
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,629	1,629	1,629	1,374
Pseudo R^2 / R^2	0.67937	0.67709	0.67943	0.68398

Note. This model is estimated by OLS. Controls include firm-level controls (size, leverage, tangibility, profitability, current ratio, market-to-book), other remaining loan terms, and loan-level controls (number of lenders, performance pricing grid, tranche type, purpose). Full coefficients, including all controls, are reported in Appendix [Tables 29 (Gap 1) through 32 (Gap 4)]. Standard errors, clustered at the firm level, are shown in parentheses.

*p < .10. ** p < .05. *** p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

The moderating role of the gap between carbon intensity and maturity is the most robust one, as the effect is negative and statistically significant across all gap measures, indicating that the maturities in two analyzed groups are in fact statistically different. For gaps 1 and 2, additionally effects for non-greenwashers are negative and statistically significant (-0.017, $p < .10$). Since effect for greenwashers is given by $\beta_1 + \beta_3$, it is -0.065 under Gap 1 measure and -0.047 under Gap 2 measure. Under Gap 3 and Gap 4 measures, the effect is reduced to interaction terms and equals -0.048 and -0.055, respectively. This effect is robust across all specifications, and therefore it can be concluded that for greenwashing companies, higher carbon intensity is associated with shorter maturities. This is consistent with H2 hypothesis. Moreover, under Gap 3 and Gap 4 measures, gaps themselves have negative and significant effects (-0.184, $p < .01$; -0.223, $p < .05$; respectively), which can be interpreted that greenwashing firms obtain shorter maturities, regardless of their carbon intensity.

Table 14

Moderating Effect of the Intention-Action Gaps on the Relationship Between Carbon Intensity and Collateral Probability

Variable	Gap 1	Gap 2	Gap 3	Gap 4
ln(CI) β_1	0.152 (0.100)	0.158 (0.108)	0.060 (0.124)	0.032 (0.107)
Gap β_2	-0.911 (0.664)	-1.70** (0.729)	-0.458 (0.535)	0.983 (1.03)
ln(CI) × Gap β_3	**-0.081 (0.211)**	**-0.390** (0.198)**	**-0.328** (0.137)**	**0.282 (0.212)**
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,629	1,629	1,629	1,374
Pseudo R^2 / R^2	0.54395	0.54782	0.55183	0.56757

Note. This model is estimated by logistic regression. Controls include firm-level controls (size, leverage, tangibility, profitability, current ratio, market-to-book), other remaining loan terms, and loan-level controls (number of lenders, performance pricing grid, tranche type, purpose). Full coefficients, including all controls, are reported in Appendix [Tables 29 (Gap 1) through 32 (Gap 4)]. Standard errors, clustered at the firm level, are shown in parentheses.

*p < .10. ** p < .05. *** p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

The relationship between carbon intensity and collateral probability is moderated by Gap 2 and Gap 3 measures. In both cases, carbon intensity coefficients β_1 for non-greenwashers are not statistically significant and thus cannot be statistically distinguished from zero. Interaction terms are negative and significant (-0.390, $p < .05$; -0.328, $p < .05$; respectively), therefore the effects for greenwashers resulting from the sum of both coefficients are attributable to just interaction terms in both cases. Therefore, it can be stated that for firms with intention-action gaps, measured with Gap 2 and Gap 3, increase in carbon intensity is associated with lower probability of loans being secured with collateral. This is contradictory with the predicted direction of the moderation, as intention-action gap results in lower probability of collateral, instead of higher. This can be interpreted based on the idea that maturity and collateral (or other monitoring-related terms) can act as substitutes, while designing loan contracts. This means that banks, after offering shorter maturities for greenwashing companies, may find it redundant to additionally impose collateral obligation in the loan. This results in probability of secured loans to decrease under this assumption. Ortiz-Molina and Penas (2008) investigated the substitution effect and found that loan maturity is longer when collateral is pledged. They interpret it as evidence that both terms act as substitute mechanisms in controlling risk. They state that collateral inclusion allows banks to grant longer maturity, as when collateral is pledged, banks do not need to rely on shorter maturities to limit their risk exposure. This shows that both terms move rather in the same direction. Applying this logic in reverse, the absence of collateral would result in shorter maturity, which is consistent with findings for greenwashers. Additionally, borrowers seem to trade off in-person or information provision monitoring with monitoring through financial covenants and loan's maturity (Gustafson et al., 2021). This suggests that shortening loan's maturity can substitute for other monitoring loan terms, although collateral inclusion is not analyzed. However, this pattern is contradictory to a substantial body of prior literature that treats price and non-price loan terms as complements to mitigate banks' risk (Berger et al., 2005; Strahan, 1999). Berger et al. (2005) explicitly find that collateral and maturity are complements in dealing with information asymmetry problems, rather than substitutes. Strahan (1999) shows that banks use price and non-price terms as complements, and that small loans, with shorter maturities and collateral experience also higher spreads. Given that the result found in this study is robust only under two gaps measures and differs from the complements view of price and non-price terms, it should be interpreted with caution. Since there are two contradictory streams in the literature, the result additionally highlights the prominent research area, which should be further explored. This may allow researchers to determine whether shorter maturities substitute for collateral as a risk-control measure for firms with environmental intention-action gap.

Table 15

Moderating Effect of the Intention-Action Gaps on the Relationship Between Carbon Intensity and Covenants Probability

Variable	Gap 1	Gap 2	Gap 3	Gap 4
ln(CI) β_1	-0.080 (0.076)	-0.034 (0.073)	0.040 (0.082)	0.044 (0.072)
Gap β_2	0.725* (0.364)	0.271 (0.408)	-0.328 (0.382)	0.320 (0.680)
ln(CI) × Gap β_3	**0.093 (0.092)**	**0.026 (0.100)**	**-0.047 (0.092)**	**-0.018 (0.139)**
Controls	**Yes**	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,629	1,629	1,629	1,374
Pseudo R^2 / R^2	0.28278	0.27974	0.28075	0.26804

Note. This model is estimated by logistic regression. Controls include firm-level controls (size, leverage, tangibility, profitability, current ratio, market-to-book), other remaining loan terms, and loan-level controls (number of lenders, performance pricing grid, tranche type, purpose). Full coefficients, including all controls, are reported in Appendix [Tables 29 (Gap 1) through 32 (Gap 4)]. Standard errors, clustered at the firm level, are shown in parentheses.

*p < .10. ** p < .05. *** p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

The relationship between carbon intensity and probability of covenants in loan contracts is not moderated by any of the gap measures. All effects are statistically not significant and cannot be distinguished from zero. Gap 1 itself has positive and significant effect (0.725, $p < .10$), indicating that greenwashing firms are more likely to obtain covenants, regardless of their carbon intensity. Nevertheless, the effect is not replicated under any other gap measure, thus has to be interpreted with caution.

Overall, only the relationship between carbon intensity and maturity is moderated by intention-action gaps, under all gap measures. The effect is negative and significant and indicates that greenwashing companies get shorter maturities. The findings of a negative effect on loan amount and spread are robust only under one specification. Therefore, even though they follow the predicted direction, they should be interpreted with caution. No significant moderation is found for covenants probability. A negative interaction is found for collateral probability under two gap measures, indicating a lower probability of collateral inclusion for greenwashing firms, which is consistent with the view that maturity and collateral act as substitute mechanisms, rather than complements. Nevertheless, this result should be examined further as there are two contradictory streams in the current literature. Taken together, it can be concluded that hypothesis 2 is only partially supported.

5. Conclusions and limitations

The paper empirically analyzes how direct carbon emissions intensity influences loan spread, amount, maturity, and probability of inclusion of collateral and covenants, as well as how the relationship changes in the presence of the intention-action gap. Intention-action gap captures the discrepancy between a firm's stated environmental commitments and its actual emissions performance. The main finding is that higher carbon intensity is associated with shorter maturities. The effect is further amplified for greenwashing companies and is robust under all four gap measures. The main result implies that banks appear to respond to carbon risk by shortening loan maturity, rather than adjusting its price.

Only one hypothesis is supported (H1c) as other loan terms provide no significant effects in the full sample. Subsample analysis reveals that the effects are hidden in subgroups and demonstrates that effects of spread and maturities are concentrated after introduction of Paris Agreement and among low profitability firms. Effect of loan amount is concentrated in brown sectors, with companies there obtaining significantly smaller loans, and companies with targets obtain shorter maturities with increase of their carbon intensity. Hypothesis 2 is only partially supported, as only relationship between carbon intensity and maturity is moderated consistently under all four gap measures. The results for remaining loan terms have either weak moderation or are inconsistent across models. Results for maturity are in line with previous findings (Chen et al., 2021; Ding et al., 2023; Ho and Wong, 2023). However, this study finds weak spread evidence, in contrast to previous studies that focus heavily on spread (Ge et al., 2025; Li, Wen and Yang, 2025; Umar et al., 2025; Zhou et al., 2025; Zhu and Zhao, 2022). This suggests that in this sample, banks' response to carbon risk is manifested through the structure of the loan (specifically maturity), rather than price itself. It shows the area for future research by implying that current studies, focusing on loan spread, may understate how lenders actually respond to carbon risk. Additionally, negative interaction is found for collateral probability under two gap measures, which combined with negative effect on maturity is in opposition to previous literature which treated collateral and maturity as complements (Berger et al., 2005; Strahan, 1999). Delegated monitoring theory (Diamond, 1984) predicts that banks in fact act as delegated monitors, as they possess an information advantage over arm's-length investors. The results show that this monitoring advantage in response to carbon-related and greenwashing-related risk is concentrated in shortening loan maturity, rather than adjusting all loan terms. The negative spread coefficient found for low profitability companies is in line with Stiglitz and Weiss's (1981) prediction that lenders may in fact avoid raising prices for risky borrowers, not to worsen adverse selection. In the sample analyzed, lenders rely on shortening maturity, rather than increasing loan prices, consistent with the theory proposed by Stiglitz and Weiss (1981). The consistent result of shorter maturities for companies with intention-action gaps suggests that lenders in fact can distinguish real environmentalism from greenwashing. This is in line with Spence's (1973) general idea that sufficiently informed party can detect an unverifiable signal over time. However, due to absence of significant results for other loan terms, this suggests that the penalty operates through a much narrower channel than the literature on greenwashing would have predicted.

This study offers several practical implications to banks, firms and regulators. For banks, the findings suggest that screening based solely on disclosure may be insufficient to distinguish credible commitments from symbolic ones. Lenders could therefore benefit from deeper incorporation of emissions-performance verification, rather than just target disclosure, while adjusting loan terms. For regulators, limited evidence of a price penalty of greenwashing implies that disclosure regulations may be insufficient and may further amplify the greenwashing problem. If lenders do not consistently price the intention-action gap into loan spread, symbolic environmentalism remains, and regulatory efforts may be needed to shift focus towards verification and actual enforcement mechanisms. For firms, the results show that publicly stating environmental commitments is not sufficient to obtain favorable loan terms. Firms whose emissions remain above their sector peers despite stated targets face significantly shorter maturities.

Nevertheless, the study experiences several limitations regarding data itself, measurement and methodology, which future research could address. First, the dataset consists of only 1,921 loan facilities issued to 425 U.S. companies, which limits the generalizability to other countries and regions not tested. Future studies could repeat the tests using a cross-country dataset, or loan facilities issued to companies in countries other than the U.S. Additionally, different sample sizes are used for distinct gap measures, due to significantly smaller coverage for Gap 4, mostly because of 630 missing observations for variable sdg13. Similarly, single-firm cells are dropped from Gap 1, Gap 2 and Gap 3, as the measures rely on sector-year medians. The exclusion may result in noisy benchmarks either way, leaving other low populated cells. Further studies should therefore introduce the dataset with fuller SDG disclosure. Intention-action gap measures are constructed using binary indicators which may miss important distinctions among firms, compared to continuous measures. Moreover, gap measures show inconsistent results on non-maturity terms, which highlights the measurement sensitivity. Thus, future research could test alternative continuous or index-based gap measures, to further examine the moderating role. Finally, used regression models show correlation and not cause, which means that the results can be only interpreted as correlations among variables and not the real cause for worse loan terms. Reverse causality is not fully ruled out, as firms obtaining worse loan terms may cut their capital expenditure, including investments in emissions reduction, which in turn results in higher carbon intensity. Lastly, the question regarding maturity and collateral relationship remains unresolved. This presents a promising line of research that should be analyzed to determine the substitution or complements mechanism, as the evidence remains mixed in the literature.

Overall, this study shows that carbon intensity shapes the loan terms in a narrower way than prior literature suggests. Carbon risk is not priced broadly across spread, amount, collateral and covenants, it is managed through the length of the loan itself. The discrepancy between what firms claim and what they actually do further results in the shorter loans. It may appear that banks do not reprice carbon risk, to respond to it they simply lend for less time.

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Appendix

Table 16

Variables definitions

Variable	Definition	Source
CI	Carbon intensity, measured as scope-1 (direct) corporate carbon emissions divided by sales revenue.	Own elaboration based on Refinitiv
spread	Total fee over base rate paid by borrower to lender on all drawn loan commitments, measured in all-in-spread-drawn (bps).	DealScan
amount	Committed funds available via individual tranche.	DealScan
maturity	Period of time (measured in months) between tranche active date and tranche maturity date.	DealScan
secured	A binary indicator determining whether the tranche is secured by collateral.	DealScan
covenants	A binary indicator determining whether financial covenants exist according to the terms and conditions of the loan.	DealScan
ta	Total assets of the company.	Refinitiv
size	The natural logarithm of firm's total assets.	Own elaboration based on Refinitiv
leverage	The ratio of total debt to total assets.	Own elaboration based on Refinitiv
tangibility	The proportion of tangible assets to total assets.	Own elaboration based on Refinitiv
profitability	Return on assets (ROA), calculated as the ratio of net income to total assets.	Own elaboration based on Refinitiv
curratio	The ratio of current assets to current liabilities.	Own elaboration based on Refinitiv
mtb	Market to book value ratio, the ratio of market value of equity of the company to its book value.	Refinitiv
numberoflenders	Number of lenders participating in the loan.	DealScan
performancepricinggrid	Indication of existence of performance-based grid used to set margin pricing on a loan.	DealScan
tranchetype	The specific type of loan construct defined by characteristics including seniority, tenor etc. marketed to different lenders/investors	DealScan
purpose	The primary purpose of the tranche.	DealScan
esg	ESG score of the company.	Refinitiv
sdg13	A binary indicator determining if company supports the UN Sustainable Development Goal 13 (SDG 13) Climate Action.	Refinitiv
excompensation	A binary indicator determining if the senior executive's compensation is linked to CSR/H&S/Sustainability targets.	Refinitiv
csr committee	A binary indicator determining if the company has a CSR committee or team.	Refinitiv

Variable	Definition	Source
target	A binary indicator determining if the company has an emissions reduction target.	Refinitiv
Gap 1	A binary indicator equal to one if the firm has a stated emissions-reduction target but its scope-1 carbon intensity is above the sector-year median.	Own elaboration based on Refinitiv
Gap 2	A binary indicator equal to one if the firm's ESG score is above the sector-year median but its carbon intensity is also above the sector-year median.	Own elaboration based on Refinitiv
Gap 3	Binary indicator equal to one if the firm has a CSR committee but its carbon intensity is above the sector-year median.	Own elaboration based on Refinitiv
Gap 4	Binary indicator equal to one if the firm supports UN SDG 13 (Climate Action) but does not have senior executive compensation linked to sustainability, CSR, or health and safety targets.	Own elaboration based on Refinitiv

Table 17

Correlation matrix among variables with significance levels

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1.ln(CI)	–													
2. ln(Spread)	.16***	–												
3. ln(Amount)	-.12***	-.26***	–											
4. ln(Maturity)	.01	.25***	-.21***	–										
5. ln(Lenders)	-.07***	-.04*	.28***	.21***	–									
6. Size	.04	-.34***	.61***	-.34***	.24***	–								
7. Leverage	.10***	.11***	.07***	-.01	.01	.05**	–							
8. Tangibility	.70***	.12***	-.11***	.01	-.12***	.10***	.12***	–						
9. Profitability	-.08***	-.14***	.04*	.00	.01	-.12***	-.03	-.07***	–					
10. Current ratio	-.19***	.10***	-.12***	.10***	-.02	-.35***	-.12***	-.33***	.06***	–				
11. Market-to-book	-.17***	-.14***	.05**	.01	.02	-.06***	.03	-.11***	.14***	.01	–			
12. Secured	.07***	.53***	-.20***	.35***	-.08***	-.32***	.13***	.02	-.04*	.07***	-.06***	–		
13. Covenants	-.03	-.01	-.02	.07***	.19***	-.04	-.09***	-.03	.03	.05**	.03	-.05**	–	
14. Performance pricing grid	-.12***	-.13***	.13***	.02	.29***	.11***	-.12***	-.12***	.06***	.03	.08***	-.22***	.50***	–

Note. ln(CI) is the independent variable. ln(Spread), ln(Amount), ln(Maturity), Secured, and Covenants are dependent variables. Size, Leverage, Tangibility, Profitability, Current ratio, and Market-to-book are firm-level control variables. Lenders and Performance pricing grid are loan-level control variables.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 18

H1a-H1e regression results with Carbon Intensity

	h1a	h1b	h1c	h1d	h1e
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.004 (0.013)**	**-0.027 (0.019)**	**-0.019** (0.010)**	**0.085 (0.097)**	**-0.011 (0.068)**
Size	-0.072*** (0.017)	0.540*** (0.025)	-0.119*** (0.017)	-0.115 (0.186)	-0.202* (0.121)
Leverage	0.237 (0.167)	0.445*** (0.158)	-0.075 (0.080)	-0.604 (1.00)	-0.603 (0.606)
Tangibility	0.051 (0.124)	-0.159 (0.179)	0.040 (0.099)	-0.486 (1.01)	0.821 (0.734)
Profitability	-0.997* (0.581)	1.13*** (0.285)	-0.245 (0.240)	-1.48 (1.63)	-0.592 (1.21)
Current ratio	0.017 (0.019)	0.042 (0.033)	-0.002 (0.023)	-0.386** (0.185)	0.104 (0.131)
Market-to-book	-0.010** (0.004)	0.011** (0.004)	0.0009 (0.002)	-0.043 (0.037)	0.003 (0.019)
ln(Amount)	-0.060*** (0.021)	–	0.031* (0.018)	-0.171 (0.166)	0.010 (0.103)
ln(Maturity)	0.049 (0.030)	0.090* (0.053)	–	2.51*** (0.577)	-0.316** (0.155)
Secured	0.363*** (0.056)	-0.070 (0.099)	0.275*** (0.070)	–	0.440 (0.353)
Covenants	0.008 (0.030)	0.006 (0.056)	-0.057** (0.029)	0.626 (0.399)	–
ln(Lenders)	0.072*** (0.022)	0.173*** (0.038)	0.168*** (0.026)	-0.199 (0.296)	0.236 (0.133)
Performance pricing grid (yes)	0.016 (0.040)	0.066 (0.058)	0.024 (0.031)	-0.271 (0.371)	2.55*** (0.200)
Tranche type: Term Loan A	0.048 (0.033)	-0.330*** (0.078)	-0.385*** (0.046)	0.029 (0.247)	0.228 (0.246)
Tranche type: Institutional term loan	0.425*** (0.053)	0.359*** (0.130)	0.074 (0.071)	3.60*** (0.910)	-0.072 (0.306)
Tranche type: Term loan (generic)	0.050 (0.050)	-0.336*** (0.109)	-0.385*** (0.077)	-0.612 (0.491)	0.317 (0.272)
Tranche type: Delay-draw term loan	-0.013 (0.045)	-0.198** (0.094)	-0.527*** (0.059)	-0.670 (0.455)	0.414 (0.284)
Tranche type: 364-day facility	-0.111 (0.077)	-0.325*** (0.094)	-1.25*** (0.036)	1.18 (1.15)	-0.370 (0.334)
Tranche type: Bridge loan	0.277*** (0.064)	0.906*** (0.136)	-1.41*** (0.053)	1.23 (1.07)	-1.10** (0.450)
Tranche type: Other	0.035 (0.155)	-0.679 (0.466)	0.120 (0.500)	-1.83 (1.95)	-14.3*** (0.312)
Purpose: Acquisition	0.044 (0.041)	0.202** (0.085)	0.021 (0.058)	0.313 (0.514)	-0.101 (0.347)
Purpose: Other	0.190*** (0.051)	0.255*** (0.071)	0.059 (0.076)	1.25*** (0.444)	-0.243 (0.319)
Purpose: Takeover	0.073* (0.037)	0.243*** (0.073)	0.003 (0.041)	0.254 (0.434)	-0.328 (0.317)
ln(Spread)	–	-0.212*** (0.062)	0.059 (0.037)	3.27*** (0.439)	0.043 (0.205)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.420	0.562	0.675	0.552	0.335
Pseudo R^2	0.353	0.270	0.486	0.540	0.273
BIC	1,973.6	4,035.8	2,280.7	1,026.3	1,966.1

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models h1a-h1c are estimated by OLS; h1d-h1e are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 19

Regression Results with Firm Size (Large) Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.008 (0.013)**	**-0.013 (0.019)**	**-0.013 (0.011)**	**0.076 (0.100)**	**-0.051 (0.072)**
Large	**-0.052 (0.071)**	**-0.005 (0.111)**	**-0.074 (0.058)**	**0.037 (0.608)**	**0.381 (0.412)**
ln(CI) × Large	**-0.008 (0.013)**	**-0.030 (0.022)**	**-0.013 (0.011)**	**0.027 (0.124)**	**0.080 (0.076)**
Size	-0.066*** (0.023)	0.503*** (0.037)	-0.112*** (0.022)	-0.095 (0.226)	-0.220 (0.165)
Leverage	0.237 (0.169)	0.458*** (0.157)	-0.073 (0.080)	-0.612 (1.01)	-0.613 (0.598)
Tangibility	0.055 (0.126)	-0.172 (0.179)	0.045 (0.100)	-0.474 (1.01)	0.811 (0.745)
Profitability	-1.01* (0.580)	1.11*** (0.288)	-0.262 (0.244)	-1.47 (1.61)	-0.494 (1.21)
Current ratio	0.019 (0.019)	0.047 (0.033)	0.0007 (0.023)	-0.388** (0.188)	0.089 (0.133)
Market-to-book	-0.010** (0.004)	0.011** (0.004)	0.001 (0.002)	-0.043 (0.037)	0.002 (0.018)
ln(Amount)	-0.060*** (0.021)	–	0.030 (0.018)	-0.171 (0.168)	0.015 (0.103)
ln(Maturity)	0.048 (0.030)	0.089* (0.053)	–	2.50*** (0.575)	-0.320** (0.155)
Secured	0.362*** (0.056)	-0.073 (0.098)	0.273*** (0.070)	–	0.445 (0.357)
Covenants	0.009 (0.030)	0.007 (0.056)	-0.056* (0.029)	0.632 (0.399)	–
ln(Lenders)	0.073*** (0.022)	0.177*** (0.038)	0.169*** (0.025)	-0.203 (0.297)	0.229* (0.133)
Performance pricing grid (yes)	0.016 (0.040)	0.062 (0.058)	0.024 (0.031)	-0.275 (0.372)	2.55*** (0.201)
Tranche type: Term Loan A	0.046 (0.033)	-0.329*** (0.077)	-0.386*** (0.046)	0.029 (0.247)	0.232 (0.248)
Tranche type: Institutional term loan	0.425*** (0.053)	0.355*** (0.128)	0.073 (0.071)	3.61*** (0.913)	-0.071 (0.310)
Tranche type: Term loan (generic)	0.051 (0.050)	-0.334*** (0.109)	-0.384*** (0.078)	-0.612 (0.492)	0.307 (0.271)
Tranche type: Delay-draw term loan	-0.017 (0.045)	-0.203** (0.091)	-0.532*** (0.059)	-0.672 (0.462)	0.434 (0.285)
Tranche type: 364-day facility	-0.113 (0.078)	-0.339*** (0.092)	-1.25*** (0.036)	1.18 (1.14)	-0.354 (0.328)
Tranche type: Bridge loan	0.273*** (0.063)	0.897*** (0.133)	-1.42*** (0.053)	1.23 (1.08)	-1.09** (0.451)
Tranche type: Other	0.024 (0.162)	-0.682 (0.466)	0.104 (0.494)	-1.81 (1.97)	-14.3*** (0.345)
Purpose: Acquisition	0.043 (0.042)	0.195** (0.084)	0.019 (0.058)	0.308 (0.517)	-0.091 (0.349)
Purpose: Other	0.191*** (0.051)	0.255*** (0.071)	0.060 (0.076)	1.25*** (0.445)	-0.254 (0.319)
Purpose: Takeover	0.072* (0.037)	0.239*** (0.073)	0.001 (0.041)	0.254 (0.435)	-0.312 (0.319)
ln(Spread)	–	-0.212*** (0.063)	0.057 (0.037)	3.27*** (0.437)	0.050 (0.205)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.42094	0.57067	0.67540	0.55161	0.33671
Pseudo R^2	0.35343	0.27276	0.48682	0.54055	0.27435
BIC	1,987.1	4,045.1	2,293.1	1,041.0	1,978.5

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS; models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 20

Regression Results with Post-Paris Agreement Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.028 (0.018)**	**-0.038 (0.025)**	**-0.0004 (0.012)**	**0.084 (0.117)**	**-0.068 (0.100)**
ln(CI) × Post-Paris	**-0.034** (0.014)**	**0.015 (0.019)**	**-0.026** (0.012)**	**0.002 (0.104)**	**0.081 (0.092)**
Size	-0.074*** (0.017)	0.541*** (0.025)	-0.121*** (0.017)	-0.115 (0.186)	-0.197 (0.123)
Leverage	0.228 (0.168)	0.448*** (0.158)	-0.080 (0.080)	-0.604 (1.00)	-0.580 (0.617)
Tangibility	0.058 (0.124)	-0.162 (0.179)	0.045 (0.098)	-0.487 (1.01)	0.793 (0.727)
Profitability	-1.04* (0.576)	1.15*** (0.282)	-0.281 (0.244)	-1.48 (1.65)	-0.483 (1.19)
Current ratio	0.016 (0.019)	0.043 (0.033)	-0.003 (0.024)	-0.386** (0.187)	0.103 (0.131)
Market-to-book	-0.010** (0.004)	0.011** (0.004)	0.0010 (0.002)	-0.043 (0.037)	0.003 (0.019)
ln(Amount)	-0.059*** (0.021)	–	0.032* (0.018)	-0.171 (0.167)	0.006 (0.103)
ln(Maturity)	0.043 (0.031)	0.092* (0.053)	–	2.51*** (0.576)	-0.302* (0.158)
Secured	0.364*** (0.056)	-0.072 (0.099)	0.277*** (0.070)	–	0.433 (0.354)
Covenants	0.011 (0.030)	0.004 (0.056)	-0.055* (0.029)	0.626 (0.400)	–
ln(Lenders)	0.072*** (0.022)	0.172*** (0.038)	0.167*** (0.026)	-0.199 (0.296)	0.237* (0.134)
Performance pricing grid (yes)	0.013 (0.039)	0.067 (0.058)	0.022 (0.031)	-0.271 (0.371)	2.56*** (0.200)
Tranche type: Term Loan A	0.050 (0.033)	-0.331*** (0.078)	-0.382*** (0.046)	0.029 (0.249)	0.222 (0.248)
Tranche type: Institutional term loan	0.421*** (0.053)	0.359*** (0.130)	0.073 (0.071)	3.60*** (0.911)	-0.076 (0.309)
Tranche type: Term loan (generic)	0.053 (0.049)	-0.337*** (0.109)	-0.381*** (0.077)	-0.612 (0.491)	0.311 (0.272)
Tranche type: Delay-draw term loan	-0.012 (0.045)	-0.198** (0.094)	-0.525*** (0.059)	-0.669 (0.454)	0.411 (0.284)
Tranche type: 364-day facility	-0.114 (0.077)	-0.323*** (0.094)	-1.25*** (0.036)	1.18 (1.15)	-0.356 (0.338)
Tranche type: Bridge loan	0.270*** (0.064)	0.908*** (0.136)	-1.41*** (0.053)	1.23 (1.07)	-1.08** (0.453)
Tranche type: Other	0.057 (0.154)	-0.689 (0.466)	0.136 (0.500)	-1.83 (1.95)	-14.3*** (0.304)
Purpose: Acquisition	0.037 (0.041)	0.204** (0.085)	0.016 (0.057)	0.313 (0.514)	-0.091 (0.348)
Purpose: Other	0.187*** (0.051)	0.256*** (0.071)	0.057 (0.076)	1.25*** (0.444)	-0.241 (0.320)
Purpose: Takeover	0.069* (0.038)	0.244*** (0.074)	0.0005 (0.041)	0.254 (0.433)	-0.317 (0.317)
ln(Spread)	–	-0.208*** (0.063)	0.052 (0.038)	3.27*** (0.437)	0.065 (0.206)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.42476	0.56942	0.67614	0.55166	0.33564
Pseudo R^2	0.35771	0.27182	0.48781	0.54048	0.27408
BIC	1,968.8	4,042.5	2,281.9	1,033.7	1,971.7

Note. Standard errors, clustered at the firm level, are shown in parentheses. The Post-Paris main effect is absorbed by the year fixed effects and is not separately identified. Models (1)-(3) are estimated by OLS; models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 21

Regression Results with Low-Profitability Moderation

	(1) Spread	(2) Amount	(3) Maturity	(4) Collateral	(5) Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.018 (0.014)**	**-0.012 (0.019)**	**-0.002 (0.013)**	**0.154 (0.119)**	**0.010 (0.082)**
Low profitability	**0.051 (0.066)**	**-0.264** (0.112)**	**-0.164* (0.090)**	**-0.473 (0.538)**	**-0.236 (0.351)**
ln(CI) × Low profitability	**-0.024** (0.012)**	**-0.026 (0.021)**	**-0.029** (0.015)**	**-0.105 (0.115)**	**-0.036 (0.075)**
Size	-0.077*** (0.017)	0.543*** (0.024)	-0.116*** (0.017)	-0.110 (0.184)	-0.199* (0.119)
Leverage	0.266 (0.165)	0.459*** (0.157)	-0.046 (0.081)	-0.503 (1.00)	-0.568 (0.626)
Tangibility	0.039 (0.123)	-0.161 (0.176)	0.033 (0.098)	-0.519 (1.03)	0.819 (0.731)
Profitability	-0.440 (0.651)	0.506 (0.371)	-0.433 (0.306)	-1.94 (1.90)	-0.936 (1.44)
Current ratio	0.028 (0.019)	0.039 (0.033)	0.0007 (0.022)	-0.374** (0.183)	0.104 (0.131)
Market-to-book	-0.008** (0.004)	0.010** (0.004)	0.0006 (0.002)	-0.043 (0.038)	0.002 (0.019)
ln(Amount)	-0.055*** (0.020)	–	0.027 (0.017)	-0.182 (0.163)	0.005 (0.101)
ln(Maturity)	0.048* (0.029)	0.078 (0.051)	–	2.48*** (0.568)	-0.324** (0.156)
Secured	0.349*** (0.051)	-0.070 (0.097)	0.269*** (0.066)	–	0.440 (0.352)
Covenants	0.009 (0.030)	0.002 (0.055)	-0.059** (0.029)	0.617 (0.399)	–
ln(Lenders)	0.072*** (0.021)	0.170*** (0.037)	0.166*** (0.026)	-0.190 (0.297)	0.235* (0.132)
Performance pricing grid (yes)	0.020 (0.040)	0.066 (0.057)	0.027 (0.031)	-0.244 (0.379)	2.56*** (0.199)
Tranche type: Term Loan A	0.047 (0.033)	-0.341*** (0.078)	-0.392*** (0.046)	-0.004 (0.245)	0.217 (0.246)
Tranche type: Institutional term loan	0.426*** (0.052)	0.358*** (0.129)	0.080 (0.069)	3.61*** (0.903)	-0.069 (0.306)
Tranche type: Term loan (generic)	0.057 (0.049)	-0.345*** (0.109)	-0.386*** (0.076)	-0.611 (0.490)	0.312 (0.272)
Tranche type: Delay-draw term loan	-0.009 (0.045)	-0.207** (0.090)	-0.530*** (0.059)	-0.675 (0.460)	0.410 (0.283)
Tranche type: 364-day facility	-0.108 (0.075)	-0.339*** (0.089)	-1.25*** (0.036)	1.14 (1.13)	-0.381 (0.334)
Tranche type: Bridge loan	0.271*** (0.062)	0.883*** (0.132)	-1.41*** (0.054)	1.16 (1.07)	-1.11** (0.451)
Tranche type: Other	0.076 (0.153)	-0.727* (0.436)	0.100 (0.483)	-1.96 (1.96)	-14.3*** (0.316)
Purpose: Acquisition	0.031 (0.043)	0.204** (0.084)	0.019 (0.058)	0.338 (0.526)	-0.106 (0.350)
Purpose: Other	0.179*** (0.049)	0.254*** (0.069)	0.056 (0.073)	1.23*** (0.447)	-0.242 (0.319)
Purpose: Takeover	0.069* (0.036)	0.238*** (0.074)	-0.0001 (0.041)	0.248 (0.439)	-0.333 (0.320)
ln(Spread)	–	-0.195*** (0.061)	0.059 (0.037)	3.27*** (0.443)	0.049 (0.206)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.43290	0.57270	0.67733	0.55158	0.33522
Pseudo R^2	0.36693	0.27429	0.48941	0.54137	0.27363
BIC	1,952.9	4,037.3	2,283.3	1,039.8	1,980.1

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS; models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 22

Regression Results with Brown-Sector Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.002 (0.011)**	**0.013 (0.015)**	**-0.015 (0.011)**	**-0.003 (0.089)**	**0.051 (0.067)**
Brown sector	**0.086 (0.068)**	**-0.298** (0.125)**	**-0.004 (0.060)**	**-0.707 (0.576)**	**-0.225 (0.391)**
ln(CI) × Brown sector	**-0.008 (0.023)**	**-0.081** (0.036)**	**-0.019 (0.013)**	**0.167 (0.218)**	**-0.163 (0.101)**
Size	-0.074*** (0.017)	0.532*** (0.025)	-0.116*** (0.017)	-0.113 (0.174)	-0.193 (0.117)
Leverage	0.194 (0.164)	0.366** (0.157)	-0.051 (0.077)	-0.518 (0.983)	-0.600 (0.596)
Tangibility	0.120 (0.112)	-0.238 (0.157)	0.062 (0.088)	-0.095 (0.834)	0.587 (0.648)
Profitability	-0.886 (0.567)	1.29*** (0.334)	-0.267 (0.230)	-1.23 (1.55)	-0.893 (1.18)
Current ratio	0.024 (0.018)	0.056* (0.032)	-0.013 (0.023)	-0.314* (0.188)	0.054 (0.128)
Market-to-book	-0.010** (0.004)	0.011** (0.004)	0.001 (0.002)	-0.041 (0.035)	0.004 (0.018)
ln(Amount)	-0.058*** (0.022)	–	0.027 (0.018)	-0.178 (0.157)	-0.015 (0.105)
ln(Maturity)	0.046 (0.031)	0.079 (0.053)	–	2.48*** (0.549)	-0.321** (0.154)
Secured	0.368*** (0.057)	-0.070 (0.097)	0.274*** (0.069)	–	0.420 (0.349)
Covenants	0.004 (0.030)	-0.008 (0.056)	-0.059** (0.029)	0.648 (0.400)	–
ln(Lenders)	0.076*** (0.022)	0.180*** (0.038)	0.166*** (0.026)	-0.190 (0.305)	0.234* (0.133)
Performance pricing grid (yes)	0.021 (0.040)	0.075 (0.058)	0.025 (0.031)	-0.325 (0.373)	2.56*** (0.200)
Tranche type: Term Loan A	0.049 (0.033)	-0.344*** (0.079)	-0.389*** (0.047)	0.050 (0.259)	0.238 (0.243)
Tranche type: Institutional term loan	0.423*** (0.053)	0.335*** (0.127)	0.073 (0.071)	3.63*** (0.907)	-0.032 (0.304)
Tranche type: Term loan (generic)	0.044 (0.051)	-0.354*** (0.108)	-0.386*** (0.078)	-0.593 (0.473)	0.326 (0.272)
Tranche type: Delay-draw term loan	-0.015 (0.045)	-0.207** (0.092)	-0.532*** (0.059)	-0.647 (0.468)	0.373 (0.284)
Tranche type: 364-day facility	-0.107 (0.080)	-0.345*** (0.094)	-1.26*** (0.037)	1.12 (1.14)	-0.404 (0.334)
Tranche type: Bridge loan	0.276*** (0.064)	0.880*** (0.134)	-1.41*** (0.054)	1.23 (1.06)	-1.11** (0.447)
Tranche type: Other	0.036 (0.160)	-0.695 (0.476)	0.107 (0.502)	-1.88 (1.88)	-14.2*** (0.264)
Purpose: Acquisition	0.054 (0.042)	0.223*** (0.086)	0.019 (0.058)	0.163 (0.527)	-0.106 (0.346)
Purpose: Other	0.187*** (0.052)	0.263*** (0.071)	0.061 (0.077)	1.22*** (0.426)	-0.216 (0.324)
Purpose: Takeover	0.081** (0.037)	0.244*** (0.075)	0.0002 (0.041)	0.275 (0.444)	-0.353 (0.318)
ln(Spread)	–	-0.205*** (0.066)	0.055 (0.037)	3.27*** (0.429)	0.011 (0.199)
Fixed effects:					
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.41539	0.56571	0.67456	0.55122	0.33513
Pseudo R^2	0.34726	0.26905	0.48571	0.53719	0.27298
BIC	1,958.3	4,019.5	2,252.8	1,001.7	1,937.2

Note. Standard errors, clustered at the firm level, are shown in parentheses. Division fixed effects are not included in the specification, as they would be collinear with the Brown/Clean classification, which is itself constructed from division-level median carbon intensity. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 23

Regression Results with Emissions-Target Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.013 (0.015)**	**-0.026 (0.025)**	**-0.005 (0.011)**	**0.076 (0.102)**	**-0.089 (0.083)**
Has target	**-0.056 (0.061)**	**-0.098 (0.107)**	**-0.065 (0.058)**	**-0.859 (0.553)**	**0.549 (0.429)**
ln(CI) × Has target	**-0.017 (0.011)**	**0.002 (0.021)**	**-0.027** (0.012)**	**0.034 (0.119)**	**0.138* (0.077)**
Size	-0.074*** (0.017)	0.549*** (0.026)	-0.124*** (0.018)	-0.0002 (0.193)	-0.194 (0.129)
Leverage	0.235 (0.168)	0.457*** (0.160)	-0.079 (0.080)	-0.419 (1.06)	-0.613 (0.617)
Tangibility	0.049 (0.124)	-0.176 (0.184)	0.041 (0.098)	-0.581 (1.03)	0.841 (0.717)
Profitability	-1.01* (0.576)	1.17*** (0.294)	-0.272 (0.247)	-1.20 (1.54)	-0.536 (1.17)
Current ratio	0.017 (0.019)	0.041 (0.033)	-0.002 (0.024)	-0.438** (0.188)	0.101 (0.132)
Market-to-book	-0.010** (0.004)	0.011** (0.004)	0.001 (0.002)	-0.041 (0.036)	0.002 (0.019)
ln(Amount)	-0.060*** (0.021)	–	0.032* (0.019)	-0.194 (0.165)	0.006 (0.104)
ln(Maturity)	0.045 (0.031)	0.092* (0.054)	–	2.69*** (0.569)	-0.298* (0.159)
Secured	0.364*** (0.057)	-0.079 (0.099)	0.278*** (0.070)	–	0.431 (0.356)
Covenants	0.011 (0.030)	0.006 (0.056)	-0.052* (0.029)	0.589 (0.394)	–
ln(Lenders)	0.072*** (0.022)	0.171*** (0.038)	0.168*** (0.025)	-0.173 (0.298)	0.242* (0.134)
Performance pricing grid (yes)	0.013 (0.039)	0.073 (0.058)	0.018 (0.031)	-0.182 (0.366)	2.57*** (0.202)
Tranche type: Term Loan A	0.048 (0.033)	-0.325*** (0.079)	-0.383*** (0.046)	0.031 (0.264)	0.212 (0.247)
Tranche type: Institutional term loan	0.423*** (0.053)	0.358*** (0.130)	0.072 (0.070)	3.65*** (0.904)	-0.070 (0.309)
Tranche type: Term loan (generic)	0.051 (0.050)	-0.330*** (0.110)	-0.382*** (0.078)	-0.533 (0.491)	0.307 (0.274)
Tranche type: Delay-draw term loan	-0.011 (0.045)	-0.194** (0.093)	-0.521*** (0.060)	-0.659 (0.455)	0.381 (0.284)
Tranche type: 364-day facility	-0.116 (0.078)	-0.319*** (0.096)	-1.25*** (0.036)	1.57 (1.12)	-0.340 (0.337)
Tranche type: Bridge loan	0.277*** (0.064)	0.909*** (0.136)	-1.40*** (0.053)	1.54 (1.06)	-1.12** (0.458)
Tranche type: Other	0.040 (0.160)	-0.683 (0.454)	0.128 (0.491)	-2.22 (1.88)	-14.4*** (0.354)
Purpose: Acquisition	0.040 (0.042)	0.204** (0.085)	0.014 (0.058)	0.360 (0.527)	-0.088 (0.351)
Purpose: Other	0.189*** (0.052)	0.247*** (0.072)	0.059 (0.076)	1.16*** (0.438)	-0.237 (0.322)
Purpose: Takeover	0.069* (0.037)	0.243*** (0.074)	-0.003 (0.041)	0.298 (0.453)	-0.288 (0.317)
ln(Spread)	–	-0.211*** (0.062)	0.054 (0.038)	3.37*** (0.437)	0.070 (0.204)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.42184	0.57006	0.67680	0.55642	0.33698
Pseudo R^2	0.35444	0.27230	0.48870	0.54810	0.27626
BIC	1,984.5	4,047.5	2,286.0	1,029.6	1,974.2

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 24

Robustness Check: Firm Size (Large) Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.010 (0.013)**	**-0.008 (0.021)**	**-0.017 (0.014)**	**0.100 (0.106)**	**-0.051 (0.071)**
Large	**-0.052 (0.071)**	**-0.053 (0.120)**	**-0.057 (0.079)**	**0.177 (0.586)**	**0.417 (0.406)**
ln(CI) × Large	**-0.006 (0.013)**	**-0.040* (0.023)**	**0.002 (0.014)**	**0.035 (0.126)**	**0.088 (0.074)**
Size	-0.082*** (0.024)	0.523*** (0.040)	-0.112*** (0.029)	-0.150 (0.229)	-0.174 (0.168)
Leverage	0.266 (0.172)	0.498*** (0.161)	0.059 (0.096)	-0.666 (0.974)	-0.586 (0.570)
Tangibility	0.059 (0.131)	-0.245 (0.190)	0.083 (0.130)	-0.615 (0.963)	0.802 (0.741)
Profitability	-1.05* (0.614)	0.994*** (0.324)	-0.240 (0.265)	-0.099 (1.87)	-0.365 (1.15)
Current ratio	0.025 (0.019)	0.072** (0.036)	-0.010 (0.025)	-0.374** (0.178)	0.074 (0.133)
Market-to-book	-0.011** (0.004)	0.011** (0.005)	0.003 (0.003)	-0.022 (0.035)	0.0009 (0.018)
ln(Amount)	-0.022 (0.018)	–	-0.029 (0.024)	-0.048 (0.174)	-0.100 (0.103)
ln(Maturity)	0.072*** (0.025)	-0.063 (0.051)	–	2.47*** (0.439)	-0.023 (0.120)
Secured	0.481*** (0.056)	0.049 (0.100)	0.564*** (0.080)	–	0.359 (0.334)
Covenants	-0.007 (0.031)	-0.062 (0.064)	-0.007 (0.036)	0.484 (0.359)	–
ln(Lenders)	0.046** (0.022)	0.148*** (0.035)	0.175*** (0.032)	-0.167 (0.245)	0.253** (0.122)
Performance pricing grid (yes)	-0.0001 (0.041)	0.057 (0.063)	0.018 (0.038)	-0.434 (0.341)	2.59*** (0.197)
Revolver (yes)	-0.067** (0.031)	0.256*** (0.074)	0.648*** (0.043)	-0.210 (0.230)	-0.173 (0.169)
General corporate purpose (yes)	-0.134*** (0.033)	-0.427*** (0.054)	0.069 (0.043)	-0.456 (0.300)	0.266 (0.202)
ln(Spread)	–	-0.084 (0.063)	0.126*** (0.045)	3.89*** (0.424)	-0.051 (0.192)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.382	0.501	0.496	0.520	0.325
Pseudo R^2	0.311	0.224	0.296	0.495	0.260
BIC	2,035.2	4,232.7	2,956.8	1,050.3	1,952.5

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 25

Robustness Check: Post-Paris Agreement Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.033* (0.018)**	**-0.032 (0.027)**	**-0.002 (0.016)**	**0.133 (0.113)**	**-0.074 (0.099)**
ln(CI) × Post-Paris	**-0.036** (0.014)**	**0.005 (0.020)**	**-0.020 (0.016)**	**-0.033 (0.105)**	**0.093 (0.091)**
Size	-0.092*** (0.018)	0.558*** (0.027)	-0.133*** (0.023)	-0.139 (0.187)	-0.150 (0.125)
Leverage	0.257 (0.171)	0.483*** (0.162)	0.060 (0.095)	-0.684 (0.954)	-0.550 (0.588)
Tangibility	0.063 (0.128)	-0.237 (0.189)	0.079 (0.127)	-0.598 (0.969)	0.793 (0.725)
Profitability	-1.08* (0.607)	1.02*** (0.313)	-0.260 (0.268)	-0.186 (1.87)	-0.301 (1.13)
Current ratio	0.022 (0.019)	0.066* (0.037)	-0.012 (0.026)	-0.367** (0.177)	0.093 (0.131)
Market-to-book	-0.010** (0.004)	0.012** (0.005)	0.003 (0.003)	-0.022 (0.035)	0.002 (0.018)
ln(Amount)	-0.021 (0.018)	–	-0.029 (0.023)	-0.045 (0.176)	-0.107 (0.103)
ln(Maturity)	0.069*** (0.025)	-0.064 (0.050)	–	2.47*** (0.438)	-0.015 (0.121)
Secured	0.481*** (0.056)	0.052 (0.099)	0.564*** (0.080)	–	0.350 (0.331)
Covenants	-0.005 (0.031)	-0.066 (0.064)	-0.005 (0.037)	0.480 (0.359)	–
ln(Lenders)	0.045** (0.022)	0.144*** (0.035)	0.175*** (0.032)	-0.166 (0.241)	0.263** (0.123)
Performance pricing grid (yes)	-0.003 (0.040)	0.061 (0.063)	0.015 (0.039)	-0.422 (0.339)	2.60*** (0.197)
Revolver (yes)	-0.069** (0.031)	0.252*** (0.076)	0.647*** (0.043)	-0.208 (0.230)	-0.158 (0.170)
General corporate purpose (yes)	-0.130*** (0.033)	-0.431*** (0.054)	0.071* (0.043)	-0.453 (0.299)	0.268 (0.202)
ln(Spread)	–	-0.083 (0.063)	0.123*** (0.045)	3.90*** (0.426)	-0.031 (0.193)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.386	0.499	0.496	0.520	0.323
Pseudo R^2	0.316	0.223	0.296	0.495	0.259
BIC	2,015.4	4,231.9	2,948.9	1,042.9	1,945.6

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 26

Robustness Check: Low-Profitability Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.020 (0.015)**	**-0.013 (0.019)**	**-0.002 (0.016)**	**0.151 (0.117)**	**0.013 (0.080)**
Low profitability	**0.073 (0.070)**	**-0.280** (0.110)**	**-0.180 (0.112)**	**-0.389 (0.473)**	**-0.203 (0.342)**
ln(CI) × Low profitability	**-0.021* (0.013)**	**-0.026 (0.022)**	**-0.024 (0.019)**	**-0.063 (0.109)**	**-0.035 (0.073)**
Size	-0.095*** (0.017)	0.562*** (0.026)	-0.127*** (0.023)	-0.132 (0.186)	-0.152 (0.121)
Leverage	0.293* (0.168)	0.494*** (0.160)	0.085 (0.096)	-0.651 (0.955)	-0.539 (0.598)
Tangibility	0.043 (0.127)	-0.237 (0.186)	0.069 (0.125)	-0.641 (0.975)	0.822 (0.728)
Profitability	-0.444 (0.685)	0.336 (0.352)	-0.551* (0.304)	-0.853 (2.18)	-0.664 (1.37)
Current ratio	0.034* (0.019)	0.061* (0.037)	-0.011 (0.025)	-0.362** (0.176)	0.094 (0.131)
Market-to-book	-0.009** (0.004)	0.010** (0.005)	0.002 (0.003)	-0.023 (0.036)	0.002 (0.019)
ln(Amount)	-0.017 (0.017)	–	-0.034 (0.023)	-0.053 (0.175)	-0.108 (0.101)
ln(Maturity)	0.075*** (0.024)	-0.074 (0.049)	–	2.45*** (0.431)	-0.027 (0.118)
Secured	0.465*** (0.051)	0.056 (0.098)	0.561*** (0.076)	–	0.355 (0.329)
Covenants	-0.008 (0.031)	-0.068 (0.062)	-0.009 (0.036)	0.462 (0.359)	–
ln(Lenders)	0.044** (0.022)	0.142*** (0.035)	0.173*** (0.032)	-0.154 (0.239)	0.259** (0.121)
Performance pricing grid (yes)	0.005 (0.041)	0.059 (0.062)	0.018 (0.039)	-0.400 (0.343)	2.59*** (0.196)
Revolver (yes)	-0.073** (0.030)	0.262*** (0.074)	0.651*** (0.043)	-0.198 (0.228)	-0.155 (0.168)
General corporate purpose (yes)	-0.125*** (0.032)	-0.427*** (0.055)	0.070* (0.042)	-0.445 (0.304)	0.280 (0.203)
ln(Spread)	–	-0.065 (0.061)	0.134*** (0.044)	3.93*** (0.430)	-0.050 (0.194)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.395	0.503	0.498	0.519	0.323
Pseudo R^2	0.325	0.226	0.298	0.496	0.259
BIC	2,000.1	4,225.9	2,948.9	1,049.6	1,954.8

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 27

Robustness Check: Brown-Sector Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.003 (0.011)**	**0.012 (0.016)**	**-0.010 (0.012)**	**0.024 (0.097)**	**0.056 (0.063)**
Brown sector	**0.098 (0.069)**	**-0.339*** (0.130)**	**0.040 (0.074)**	**-0.824 (0.585)**	**-0.190 (0.387)**
ln(CI) × Brown	**-0.006 (0.024)**	**-0.094** (0.037)**	**-0.014 (0.018)**	**0.149 (0.214)**	**-0.157 (0.101)**
Size	-0.092*** (0.018)	0.549*** (0.026)	-0.130*** (0.022)	-0.124 (0.173)	-0.149 (0.119)
Leverage	0.224 (0.166)	0.399** (0.161)	0.103 (0.093)	-0.563 (0.949)	-0.559 (0.567)
Tangibility	0.126 (0.116)	-0.282* (0.163)	0.040 (0.105)	-0.204 (0.790)	0.566 (0.646)
Profitability	-0.955 (0.596)	1.18*** (0.358)	-0.348 (0.248)	-0.076 (1.75)	-0.659 (1.13)
Current ratio	0.029 (0.019)	0.075** (0.036)	-0.024 (0.025)	-0.319* (0.184)	0.047 (0.127)
Market-to-book	-0.011** (0.004)	0.011** (0.005)	0.004 (0.003)	-0.019 (0.033)	0.003 (0.018)
ln(Amount)	-0.020 (0.019)	–	-0.034 (0.024)	-0.057 (0.167)	-0.124 (0.104)
ln(Maturity)	0.069*** (0.026)	-0.073 (0.051)	–	2.46*** (0.417)	-0.020 (0.118)
Secured	0.485*** (0.056)	0.052 (0.100)	0.567*** (0.080)	–	0.354 (0.326)
Covenants	-0.012 (0.031)	-0.080 (0.064)	-0.006 (0.036)	0.494 (0.356)	–
ln(Lenders)	0.050** (0.022)	0.153*** (0.035)	0.173*** (0.033)	-0.146 (0.243)	0.253** (0.122)
Performance pricing grid (yes)	0.003 (0.041)	0.069 (0.063)	0.013 (0.038)	-0.469 (0.342)	2.59*** (0.197)
Revolver (yes)	-0.069** (0.032)	0.269*** (0.074)	0.653*** (0.043)	-0.226 (0.228)	-0.151 (0.168)
General corporate purpose (yes)	-0.139*** (0.034)	-0.440*** (0.056)	0.070 (0.043)	-0.432 (0.296)	0.286 (0.201)
ln(Spread)	–	-0.078 (0.068)	0.121*** (0.046)	3.89*** (0.415)	-0.077 (0.189)
Fixed effects:					
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.377	0.497	0.493	0.518	0.323
Pseudo R^2	0.306	0.222	0.294	0.491	0.258
BIC	2,003.8	4,202.0	2,921.2	1,011.6	1,911.7

Note. Standard errors, clustered at the firm level, are shown in parentheses. Division fixed effects are not included in the specification, as they would be collinear with the Brown/Clean classification, which is itself constructed from division-level median carbon intensity. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 28

Robustness Check: Emissions-Target Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable:	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.016 (0.015)**	**-0.028 (0.026)**	**-0.002 (0.013)**	**0.106 (0.097)**	**-0.090 (0.082)**
Has target	**-0.061 (0.062)**	**-0.104 (0.117)**	**-0.069 (0.070)**	**-0.806 (0.525)**	**0.596 (0.433)**
ln(CI) × Has target	**-0.016 (0.011)**	**0.003 (0.023)**	**-0.026** (0.013)**	**0.020 (0.112)**	**0.144* (0.077)**
Size	-0.092*** (0.018)	0.568*** (0.027)	-0.136*** (0.023)	-0.039 (0.188)	-0.149 (0.130)
Leverage	0.266 (0.171)	0.495*** (0.164)	0.061 (0.095)	-0.524 (0.988)	-0.593 (0.586)
Tangibility	0.052 (0.129)	-0.256 (0.195)	0.076 (0.126)	-0.749 (0.969)	0.847 (0.713)
Profitability	-1.05* (0.610)	1.05*** (0.320)	-0.257 (0.266)	0.056 (1.79)	-0.351 (1.10)
Current ratio	0.023 (0.019)	0.065* (0.037)	-0.011 (0.026)	-0.399** (0.177)	0.089 (0.131)
Market-to-book	-0.011** (0.004)	0.011** (0.005)	0.003 (0.003)	-0.023 (0.035)	0.001 (0.018)
ln(Amount)	-0.022 (0.018)	–	-0.029 (0.024)	-0.054 (0.174)	-0.106 (0.105)
ln(Maturity)	0.070*** (0.025)	-0.063 (0.051)	–	2.49*** (0.437)	-0.006 (0.121)
Secured	0.481*** (0.057)	0.041 (0.099)	0.565*** (0.080)	–	0.361 (0.334)
Covenants	-0.006 (0.031)	-0.065 (0.064)	-0.003 (0.037)	0.445 (0.358)	–
ln(Lenders)	0.045** (0.022)	0.143*** (0.035)	0.175*** (0.032)	-0.141 (0.242)	0.266** (0.123)
Performance pricing grid (yes)	-0.002 (0.041)	0.069 (0.062)	0.012 (0.038)	-0.355 (0.333)	2.61*** (0.199)
Revolver (yes)	-0.068** (0.032)	0.246*** (0.076)	0.647*** (0.043)	-0.251 (0.230)	-0.155 (0.169)
General corporate purpose (yes)	-0.132*** (0.033)	-0.428*** (0.055)	0.071* (0.042)	-0.425 (0.299)	0.261 (0.202)
ln(Spread)	–	-0.083 (0.062)	0.124*** (0.046)	3.96*** (0.425)	-0.034 (0.191)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,640	1,640	1,640	1,640	1,640
R^2 / Squared cor.	0.383	0.500	0.497	0.522	0.325
Pseudo R^2	0.312	0.224	0.297	0.502	0.262
BIC	2,033.0	4,236.1	2,952.9	1,039.9	1,947.9

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 29

Hypothesis 2 Results: Regression Results with Gap 1 Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.011 (0.014)**	**-0.019 (0.022)**	**-0.017* (0.010)**	**0.152 (0.100)**	**-0.080 (0.076)**
Gap1	**-0.078 (0.065)**	**-0.107 (0.096)**	**-0.085 (0.056)**	**-0.911 (0.664)**	**0.725* (0.375)**
ln(CI) × Gap1	**-0.011 (0.017)**	**-0.028 (0.025)**	**-0.048*** (0.013)**	**-0.081 (0.211)**	**0.093 (0.092)**
Size	-0.072*** (0.017)	0.540*** (0.025)	-0.117*** (0.017)	-0.090 (0.188)	-0.208* (0.124)
Leverage	0.246 (0.169)	0.457*** (0.160)	-0.075 (0.079)	-0.527 (1.02)	-0.701 (0.625)
Tangibility	0.038 (0.125)	-0.168 (0.183)	0.059 (0.099)	-0.587 (1.01)	0.938 (0.759)
Profitability	-0.991* (0.585)	1.12*** (0.296)	-0.225 (0.236)	-1.33 (1.59)	-0.627 (1.16)
Current ratio	0.015 (0.019)	0.045 (0.033)	0.002 (0.024)	-0.438** (0.196)	0.093 (0.135)
Market-to-book	-0.010** (0.004)	0.012*** (0.004)	0.001 (0.002)	-0.036 (0.036)	0.0010 (0.019)
ln(Amount)	-0.061*** (0.021)	–	0.029 (0.018)	-0.186 (0.165)	0.006 (0.105)
ln(Maturity)	0.049 (0.030)	0.087 (0.054)	–	2.60*** (0.596)	-0.279* (0.160)
Secured	0.358*** (0.057)	-0.073 (0.099)	0.278*** (0.070)	–	0.430 (0.363)
Covenants	0.011 (0.031)	0.006 (0.057)	-0.049* (0.030)	0.628 (0.412)	–
ln(Lenders)	0.072*** (0.022)	0.172*** (0.038)	0.163*** (0.025)	-0.196 (0.293)	0.264 (0.135)
Performance pricing grid (yes)	0.015 (0.040)	0.067 (0.058)	0.020 (0.031)	-0.229 (0.376)	2.59*** (0.202)
Tranche type: Term Loan A	0.050 (0.034)	-0.328*** (0.080)	-0.391*** (0.047)	0.051 (0.248)	0.282 (0.246)
Tranche type: Institutional term loan	0.423*** (0.053)	0.357*** (0.130)	0.063 (0.069)	3.61*** (0.912)	-0.007 (0.312)
Tranche type: Term loan (generic)	0.055 (0.050)	-0.330*** (0.112)	-0.395*** (0.079)	-0.504 (0.500)	0.306 (0.281)
Tranche type: Delay-draw term loan	-0.005 (0.046)	-0.181* (0.095)	-0.536*** (0.059)	-0.632 (0.462)	0.373 (0.289)
Tranche type: 364-day facility	-0.111 (0.078)	-0.329*** (0.096)	-1.26*** (0.036)	1.38 (1.13)	-0.319 (0.345)
Tranche type: Bridge loan	0.286*** (0.065)	0.913*** (0.139)	-1.42*** (0.054)	1.49 (1.07)	-1.17** (0.466)
Tranche type: Other	0.047 (0.160)	-0.659 (0.453)	0.132 (0.487)	-1.87 (1.86)	-14.3*** (0.303)
Purpose: Acquisition	0.038 (0.043)	0.199** (0.087)	0.021 (0.058)	0.378 (0.532)	0.038 (0.329)
Purpose: Other	0.186*** (0.052)	0.248*** (0.072)	0.052 (0.076)	1.25*** (0.439)	-0.211 (0.323)
Purpose: Takeover	0.070* (0.038)	0.224*** (0.074)	-0.005 (0.041)	0.281 (0.446)	-0.268 (0.324)
ln(Spread)	–	-0.213*** (0.063)	0.058 (0.037)	3.31*** (0.448)	0.060 (0.207)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,629	1,629	1,629	1,629	1,629
Pseudo R^2 / R^2	0.42219	0.56992	0.67937	0.54395	0.28278
BIC	1,981.5	4,025.9	2,261.5	1,033.7	1,949.1

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression. R^2 values are reported for OLS models and pseudo-R^2 (McFadden) for logit models.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 30

Hypothesis 2 Results: Regression Results with Gap 2 Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.008 (0.013)**	**-0.023 (0.018)**	**-0.017* (0.010)**	**0.158 (0.108)**	**-0.034 (0.073)**
Gap2	**-0.014 (0.049)**	**-0.107 (0.123)**	**-0.057 (0.052)**	**-1.70** (0.729)**	**0.271 (0.408)**
ln(CI) × Gap2	**0.019 (0.013)**	**-0.056* (0.032)**	**-0.030** (0.013)**	**-0.390** (0.198)**	**0.026 (0.100)**
Size	-0.073*** (0.017)	0.542*** (0.025)	-0.115*** (0.017)	-0.082 (0.185)	-0.648 (0.613)
Leverage	0.253 (0.167)	0.425*** (0.160)	-0.093 (0.081)	-0.810 (1.05)	-0.648 (0.748)
Tangibility	0.038 (0.125)	-0.156 (0.176)	0.048 (0.099)	-0.533 (1.04)	0.884 (0.748)
Profitability	-0.962* (0.578)	1.05*** (0.281)	-0.257 (0.242)	-1.44 (1.83)	-0.591 (1.22)
Current ratio	0.013 (0.019)	0.050 (0.034)	0.002 (0.024)	-0.401** (0.193)	0.083 (0.134)
Market-to-book	-0.010** (0.004)	0.012*** (0.004)	0.001 (0.002)	-0.038 (0.038)	0.001 (0.019)
ln(Amount)	-0.058*** (0.021)	–	0.029 (0.018)	-0.167 (0.168)	0.004 (0.104)
ln(Maturity)	0.053 (0.030)	0.084 (0.053)	–	2.53*** (0.592)	-0.281* (0.157)
Secured	0.355*** (0.054)	-0.075 (0.099)	0.275*** (0.070)	–	0.441 (0.360)
Covenants	0.009 (0.031)	0.002 (0.057)	-0.050* (0.029)	0.637 (0.401)	–
ln(Lenders)	0.072*** (0.021)	0.174*** (0.038)	0.166*** (0.025)	-0.175 (0.302)	0.244 (0.134)
Performance pricing grid (yes)	0.015 (0.041)	0.066 (0.058)	0.021 (0.031)	-0.261 (0.374)	2.59*** (0.201)
Tranche type: Term Loan A	0.053 (0.034)	-0.334*** (0.078)	-0.392*** (0.047)	0.013 (0.236)	0.291 (0.241)
Tranche type: Institutional term loan	0.426*** (0.053)	0.358*** (0.130)	0.067 (0.071)	3.60*** (0.891)	-0.044 (0.307)
Tranche type: Term loan (generic)	0.055 (0.050)	-0.338*** (0.109)	-0.397*** (0.078)	-0.620 (0.497)	0.335 (0.276)
Tranche type: Delay-draw term loan	-0.010 (0.046)	-0.190** (0.094)	-0.540*** (0.058)	-0.696 (0.440)	0.451 (0.290)
Tranche type: 364-day facility	-0.101 (0.078)	-0.338*** (0.092)	-1.26*** (0.036)	1.20 (1.15)	-0.320 (0.340)
Tranche type: Bridge loan	0.287*** (0.065)	0.898*** (0.136)	-1.42*** (0.054)	1.27 (1.12)	-1.14** (0.467)
Tranche type: Other	0.031 (0.155)	-0.668 (0.467)	0.118 (0.493)	-1.92 (2.03)	-14.3*** (0.320)
Purpose: Acquisition	0.039 (0.042)	0.207** (0.087)	0.029 (0.058)	0.373 (0.516)	0.012 (0.327)
Purpose: Other	0.184*** (0.051)	0.251*** (0.072)	0.058 (0.075)	1.14** (0.447)	-0.203 (0.320)
Purpose: Takeover	0.073* (0.037)	0.229*** (0.074)	0.004 (0.041)	0.209 (0.439)	-0.309 (0.324)
ln(Spread)	–	-0.202*** (0.062)	0.064* (0.037)	3.40*** (0.478)	0.045 (0.207)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,629	1,629	1,629	1,629	1,629
Pseudo R^2 / R^2	0.42409	0.57150	0.67709	0.54782	0.27974
BIC	1,976.1	4,019.9	2,273.0	1,027.8	1,955.9

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression. R^2 values are reported for OLS models and pseudo-R^2 (McFadden) for logit models.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 31

Hypothesis 2 Results: Regression Results with Gap 3 Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.001 (0.014)**	**0.004 (0.021)**	**0.003 (0.011)**	**0.060 (0.124)**	**0.040 (0.082)**
Gap3	**0.059 (0.054)**	**-0.213** (0.099)**	**-0.184*** (0.062)**	**-0.458 (0.535)**	**-0.328 (0.382)**
ln(CI) × Gap3	**0.032** (0.014)**	**-0.034 (0.025)**	**-0.048*** (0.014)**	**-0.328** (0.137)**	**-0.047 (0.092)**
Size	-0.075*** (0.018)	0.540*** (0.025)	-0.113*** (0.018)	-0.161 (0.175)	-0.193 (0.124)
Leverage	0.263 (0.167)	0.426*** (0.157)	-0.107 (0.082)	-1.20 (1.06)	-0.648 (0.611)
Tangibility	0.046 (0.124)	-0.187 (0.181)	0.036 (0.097)	-0.423 (1.05)	0.825 (0.740)
Profitability	-0.990* (0.575)	1.16*** (0.295)	-0.187 (0.228)	-1.57 (1.75)	-0.443 (1.21)
Current ratio	0.013 (0.019)	0.040 (0.033)	0.002 (0.024)	-0.388* (0.212)	0.065 (0.134)
Market-to-book	-0.010** (0.004)	0.011** (0.004)	0.001 (0.002)	-0.038 (0.039)	0.003 (0.019)
ln(Amount)	-0.059*** (0.021)	–	0.028 (0.019)	-0.142 (0.152)	-0.0010 (0.104)
ln(Maturity)	0.058* (0.030)	0.083 (0.054)	–	2.46*** (0.571)	-0.291* (0.159)
Secured	0.369*** (0.057)	-0.081 (0.097)	0.266*** (0.068)	–	0.414 (0.361)
Covenants	0.009 (0.031)	-0.002 (0.057)	-0.053* (0.029)	0.548 (0.389)	–
ln(Lenders)	0.069*** (0.022)	0.178*** (0.038)	0.169*** (0.025)	-0.057 (0.295)	0.231* (0.137)
Performance pricing grid (yes)	0.019 (0.041)	0.075 (0.058)	0.020 (0.031)	-0.292 (0.371)	2.61*** (0.203)
Tranche type: Term Loan A	0.057* (0.034)	-0.324*** (0.080)	-0.393*** (0.047)	0.007 (0.246)	0.281 (0.240)
Tranche type: Institutional term loan	0.425*** (0.054)	0.365*** (0.132)	0.063 (0.072)	3.65*** (0.871)	-0.114 (0.312)
Tranche type: Term loan (generic)	0.054 (0.050)	-0.322*** (0.111)	-0.399*** (0.077)	-0.499 (0.478)	0.292 (0.277)
Tranche type: Delay-draw term loan	-0.008 (0.045)	-0.180* (0.095)	-0.536*** (0.059)	-0.619 (0.452)	0.453 (0.293)
Tranche type: 364-day facility	-0.096 (0.077)	-0.335*** (0.092)	-1.26*** (0.036)	1.16 (1.10)	-0.379 (0.342)
Tranche type: Bridge loan	0.298*** (0.065)	0.911*** (0.137)	-1.42*** (0.054)	1.14 (1.07)	-1.14** (0.477)
Tranche type: Other	0.028 (0.152)	-0.710 (0.481)	0.093 (0.474)	-1.56 (1.89)	-14.3*** (0.305)
Purpose: Acquisition	0.036 (0.043)	0.195** (0.087)	0.027 (0.058)	0.475 (0.560)	0.014 (0.331)
Purpose: Other	0.186*** (0.052)	0.234*** (0.070)	0.048 (0.073)	1.30*** (0.467)	-0.290 (0.316)
Purpose: Takeover	0.076** (0.038)	0.224*** (0.074)	-0.0008 (0.041)	0.153 (0.463)	-0.311 (0.323)
ln(Spread)	–	-0.206*** (0.063)	0.069* (0.037)	3.46*** (0.468)	0.054 (0.213)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,619	1,619	1,619	1,619	1,619
Pseudo R^2 / R^2	0.42595	0.57077	0.67943	0.55183	0.28075
BIC	1,967.6	4,000.8	2,249.5	1,014.2	1,943.2

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression. R^2 values are reported for OLS models and pseudo-R^2 (McFadden) for logit models.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.

Table 32

Hypothesis 2 Results: Regression Results with Gap 4 Moderation

	(1)	(2)	(3)	(4)	(5)
	Spread	Amount	Maturity	Collateral	Covenants
Dependent variable	*ln(Spread)*	*ln(Amount)*	*ln(Maturity)*	*Secured*	*Covenants*
ln(CI)	**0.002 (0.015)**	**-0.022 (0.019)**	**-0.008 (0.011)**	**0.032 (0.107)**	**0.044 (0.072)**
Gap4	**-0.023 (0.089)**	**0.268 (0.181)**	**-0.223** (0.091)**	**0.983 (1.03)**	**0.320 (0.680)**
ln(CI) × Gap4	**0.006 (0.017)**	**0.051 (0.034)**	**-0.055*** (0.018)**	**0.282 (0.212)**	**-0.018 (0.139)**
Size	-0.065*** (0.019)	0.520*** (0.027)	-0.119*** (0.019)	-0.020 (0.201)	-0.242* (0.127)
Leverage	0.225 (0.164)	0.432*** (0.165)	-0.062 (0.080)	-0.809 (1.13)	-0.659 (0.648)
Tangibility	0.033 (0.130)	-0.179 (0.177)	0.030 (0.104)	-1.02 (1.06)	1.11 (0.804)
Profitability	-0.819 (0.556)	1.06*** (0.307)	-0.338 (0.226)	-2.07 (1.74)	-0.847 (1.18)
Current ratio	0.018 (0.021)	0.035 (0.036)	0.007 (0.025)	-0.439* (0.231)	0.047 (0.146)
Market-to-book	-0.010** (0.004)	0.011** (0.004)	0.002 (0.002)	-0.038 (0.046)	0.009 (0.020)
ln(Amount)	-0.078*** (0.024)	–	0.017 (0.019)	-0.349** (0.139)	-0.018 (0.107)
ln(Maturity)	0.060* (0.031)	0.046 (0.055)	–	2.59*** (0.732)	-0.379** (0.174)
Secured	0.359*** (0.059)	-0.175* (0.096)	0.264*** (0.069)	–	0.411 (0.382)
Covenants	0.011 (0.032)	-0.011 (0.057)	-0.068** (0.033)	0.701 (0.472)	–
ln(Lenders)	0.074*** (0.024)	0.190*** (0.037)	0.190*** (0.026)	-0.318 (0.294)	0.254* (0.147)
Performance pricing grid (yes)	0.019 (0.040)	0.071 (0.062)	0.018 (0.033)	-0.094 (0.417)	2.52*** (0.221)
Tranche type: Term Loan A	0.043 (0.038)	-0.356*** (0.082)	-0.424*** (0.051)	-0.061 (0.296)	0.094 (0.264)
Tranche type: Institutional term loan	0.401*** (0.066)	0.457*** (0.156)	0.089 (0.074)	4.36*** (1.14)	-0.144 (0.349)
Tranche type: Term loan (generic)	0.026 (0.058)	-0.330*** (0.104)	-0.458*** (0.086)	-0.601 (0.534)	0.256 (0.302)
Tranche type: Delay-draw term loan	-0.026 (0.048)	-0.206** (0.093)	-0.542*** (0.062)	-0.522 (0.439)	0.417 (0.286)
Tranche type: 364-day facility	-0.094 (0.080)	-0.408*** (0.092)	-1.25*** (0.039)	1.10 (1.24)	-0.467 (0.358)
Tranche type: Bridge loan	0.289*** (0.067)	0.840*** (0.131)	-1.38*** (0.057)	1.84 (1.19)	-0.936* (0.480)
Tranche type: Other	0.156 (0.171)	-1.09*** (0.391)	-0.430 (0.575)	-5.15 (3.28)	-14.8*** (0.428)
Purpose: Acquisition	0.054 (0.042)	0.201** (0.089)	-0.030 (0.060)	0.249 (0.612)	-0.365 (0.398)
Purpose: Other	0.215*** (0.055)	0.252*** (0.075)	-0.002 (0.065)	1.60*** (0.486)	-0.329 (0.368)
Purpose: Takeover	0.077* (0.042)	0.208*** (0.078)	0.008 (0.040)	0.511 (0.472)	-0.252 (0.341)
ln(Spread)	–	-0.256*** (0.062)	0.071* (0.038)	3.26*** (0.523)	0.064 (0.218)
Fixed effects:					
Division	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	Logit	Logit
Observations	1,374	1,374	1,374	1,374	1,374
Pseudo R^2 / R^2	0.42349	0.58191	0.68398	0.56757	0.26804
BIC	1,739.0	3,368.5	1,966.8	865.15	1,713.7

Note. Standard errors, clustered at the firm level, are shown in parentheses. Models (1)-(3) are estimated by OLS. Models (4)-(5) are estimated by logistic regression. R^2 values are reported for OLS models and pseudo-R^2 (McFadden) for logit models.

*p < .10. **p < .05. ***p < .01.

Source: Author's own elaboration based on data from DealScan and Refinitiv.